

## Creating a Simulation Object with Animation

### Create a Simulation Object with Animation

While you are creating your simulation model, you will also model simulation objects which animate a certain work process in 3D.

In our example we model our own station in a *Frame*. We will demonstrate how to:

- [Create the Simulation Object](#)
- [Add Animations and Interactions to the Simulation Objects](#)
- [Test the Station in the Simulation Model](#)
- [Fine-tune the Station](#)

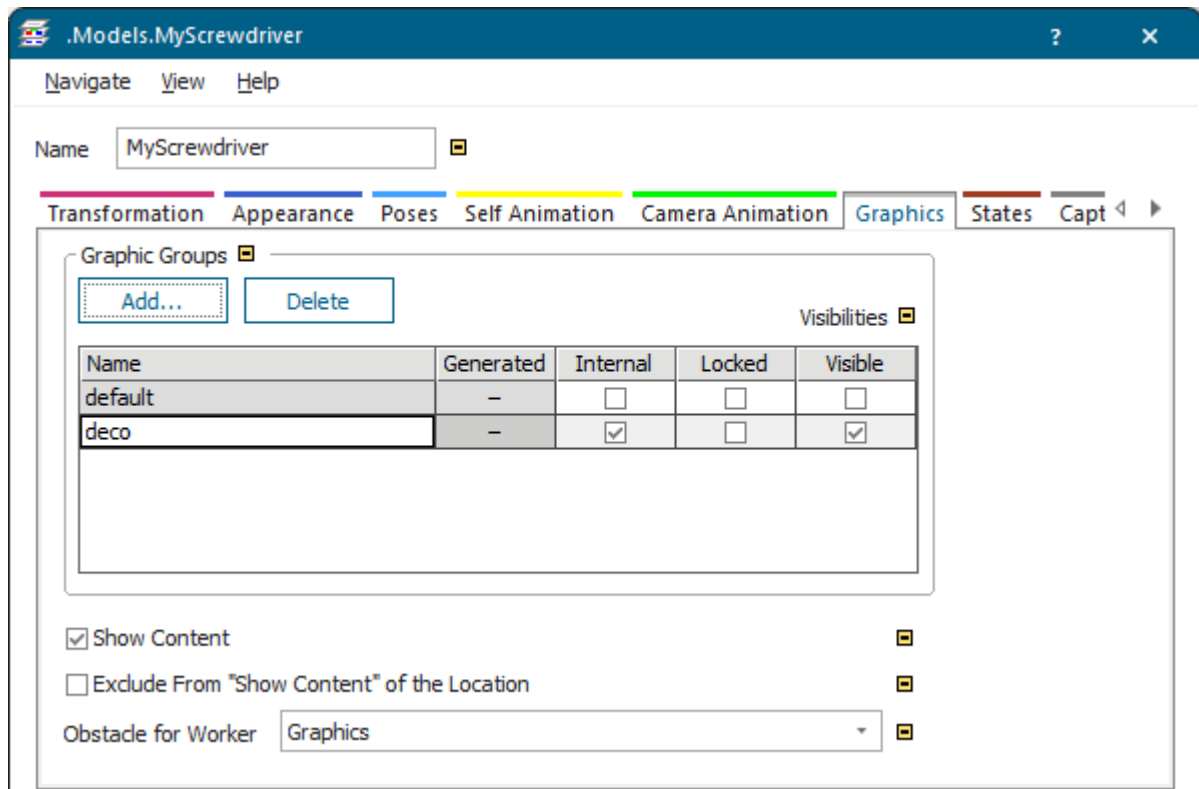
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### Create the Simulation Object

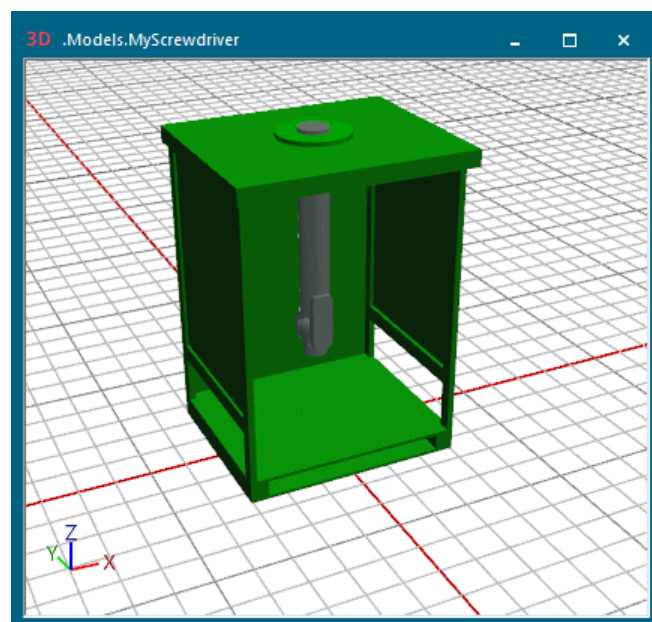
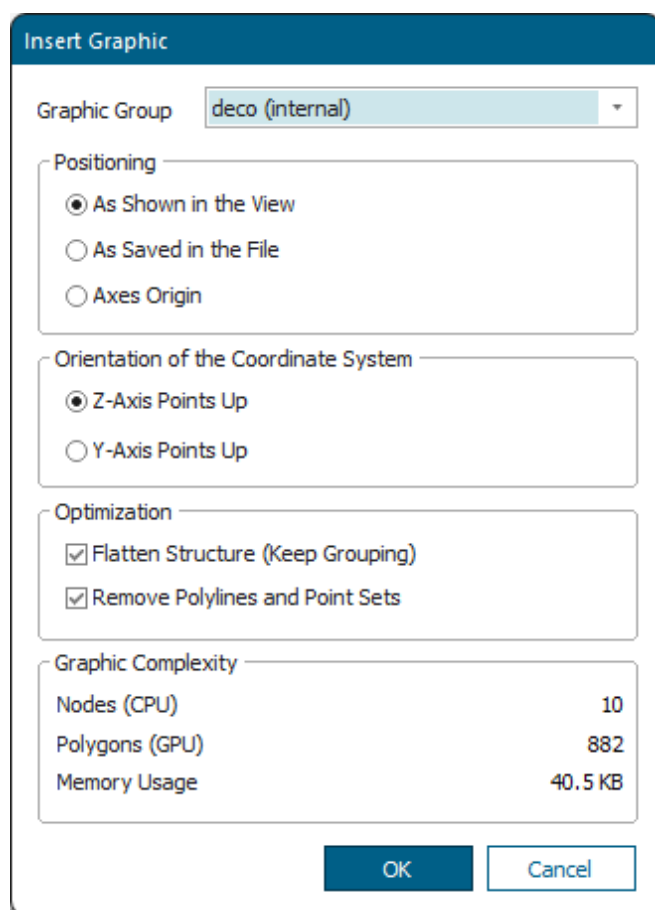
You can create the simulation object that is required for our animatable object.

Proceed as follows:

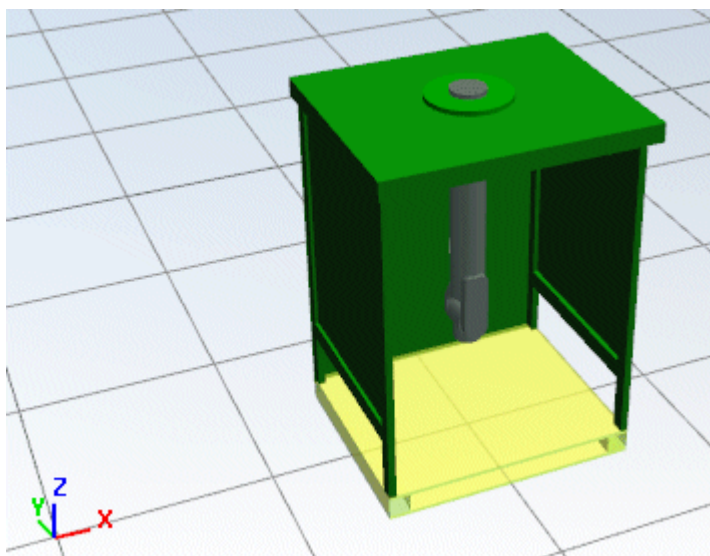
- Create a new *Frame* and **open** it. Open the dialog **Edit 3D Properties**. Change to the tab **Graphics**. Select the check box **Show Content** to show the objects contained within it.



- Click **Show External Graphic Groups**  on the **View** ribbon tab to show the **default** graphic of the *Frame*. Delete the graphic in the graphic group named **default**.
- Click **Import Graphics** on the **Edit** ribbon tab to import an appropriate JT file into the graphic group named **default** of the *Frame*. Our file is called *Screwdriver.jt*. We received it from a colleague.

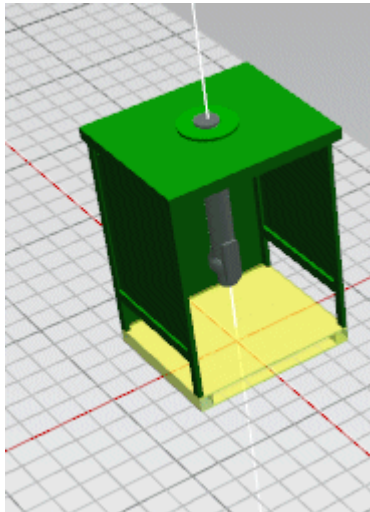


- Extract the first simulation object: Click the floor plate of the station by pressing the + key on your keyboard until only the floor plate, and nothing else, is marked as selected. 3D then shows the floor plate in a yellowish green.



- Click the graphic with the right mouse button and select **Make Simulation Object**.

- Select **.MaterialFlow.Station** as the **class**. **Name** the object **PartCarrierStation** and change the coordinates of the **Object Position** until they are located in the center of the screwdriver. Watch how the white cross lines change their position in the scene window as you change the position of the object in the dialog.



**Make Simulation Object** ? X

Name:

Class:

Object Position

X:  m

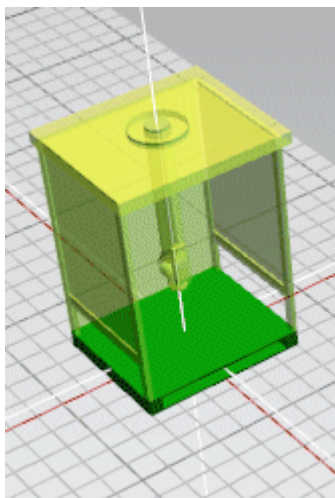
Y:  m

Z:  m

OK Cancel

- Select the trestle that holds the actual screwdriver of our screwdriver station. Notice that when you select **PartCarrierStation**, *Plant Simulation* shows it in a darker green while it shows the other graphics in a light yellowish green. **Dark green** denotes that an **object** is **selected**. **Yellowish green** denotes that a **graphic** is **selected**, meaning that you can use the + key to contract the selection along the graphic structure.

Create a second simulation object. Select **.MaterialFlow.Station** as the **class**. **Name** the object **Trestle** and change the coordinates of the **Object Position** until they are located in the center of the screwdriver.



**Make Simulation Object** ? X

Name:

Class:

Object Position

X:  m

Y:  m

Z:  m

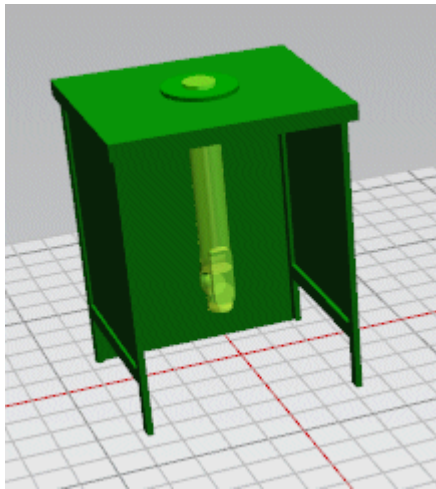
OK Cancel

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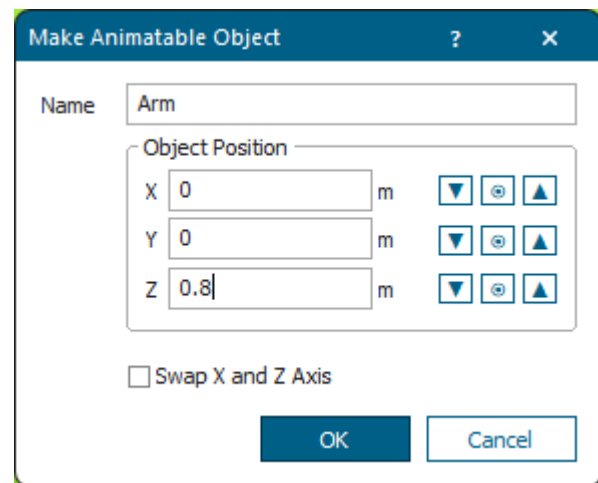
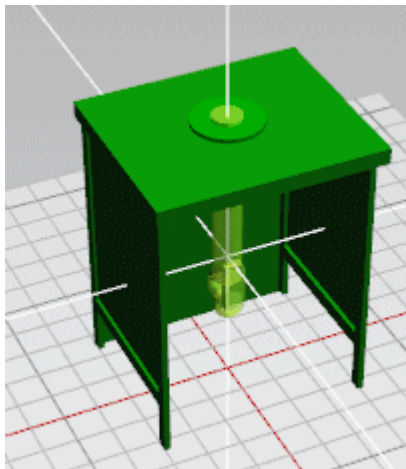
## Add Animations and Interactions to the Simulation Objects

After creating the basic simulation part of our *Screwdriver* station, we are going to add the animation part and the interaction part to it.

- Click the trestle of the screwdriver with the right mouse button and select **Open in New Window**. This opens the *Frame .Models.MyScrewdriver.Trestle*. Press the + key until only the graphic of the screw arm proper is selected, denoted by the lighter green.



- Click the graphic with the right mouse button and select the command **Make Animatable Object**. **Name** the object Arm and select the following coordinates as the **Object Position** until they are located in the center of the screw arm.

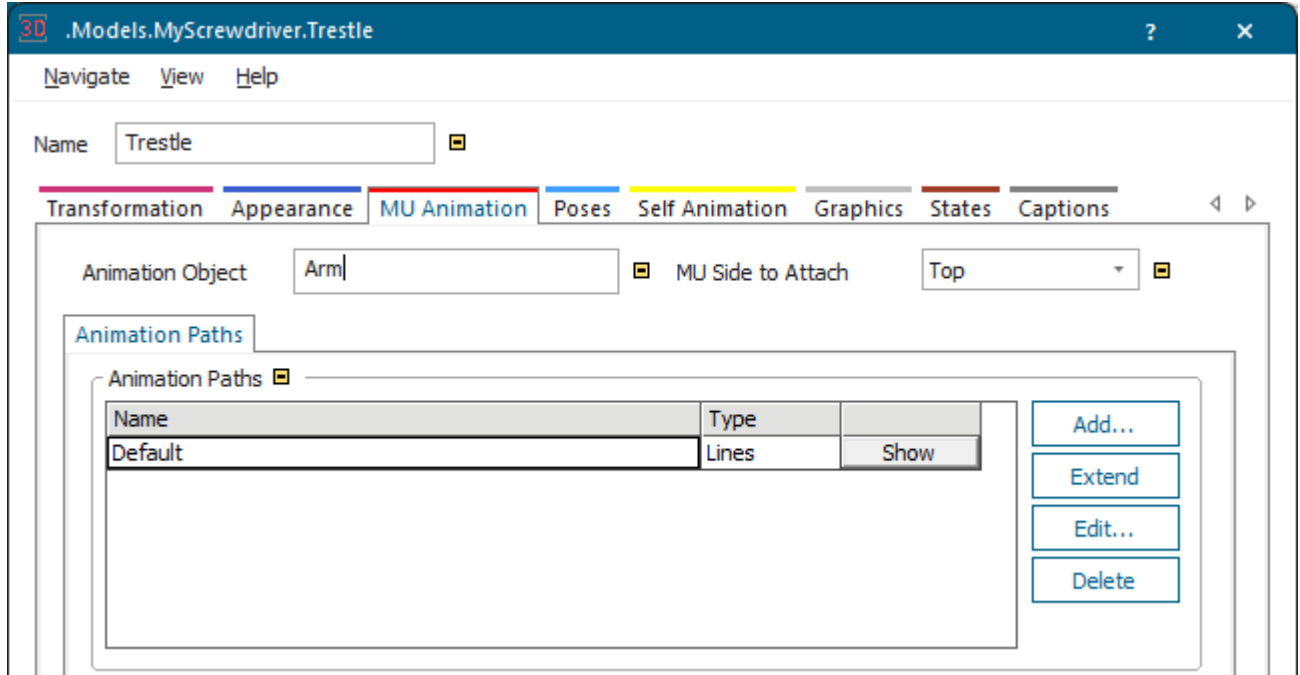


- Click the newly created animatable object with the right mouse button and select **Use as Animation Object**. This means that our trestle forwards the animation of the *MUs* to the animatable object named **Arm**.

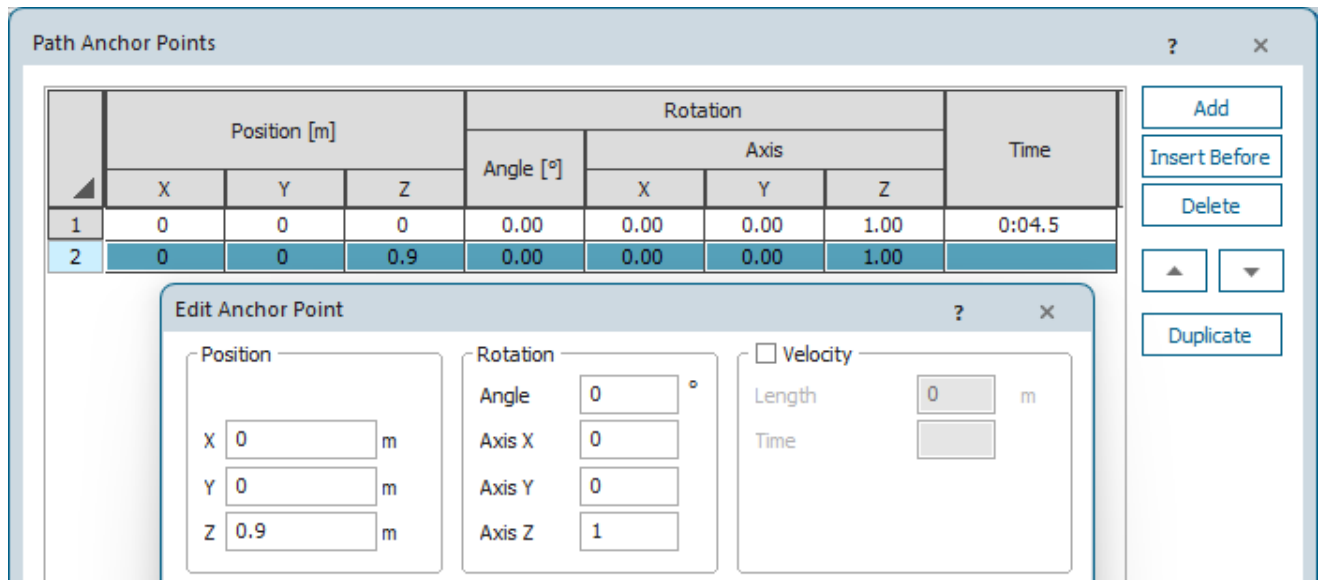
Close the *Frame .Models.MyScrewdriver.Trestle* to return to the *Frame .Models.MyScrewdriver*.

Click the **Trestle** with the right mouse button, select **Edit 3D Properties**, and change to the **MU Animation [tab]**. You'll notice that the command above entered the animation object **Arm** as the

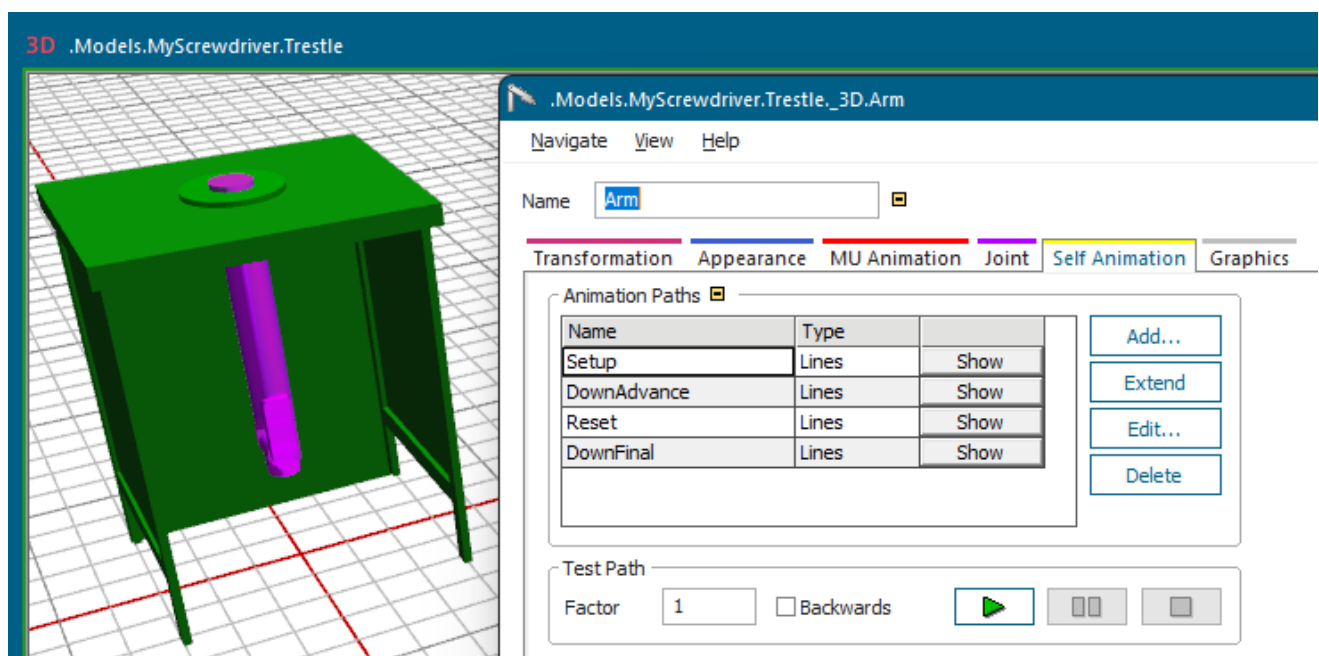
**Animation Object.** Instead of using the command **Use as Animation Object**, you could also type the name of the animation object into the text box.



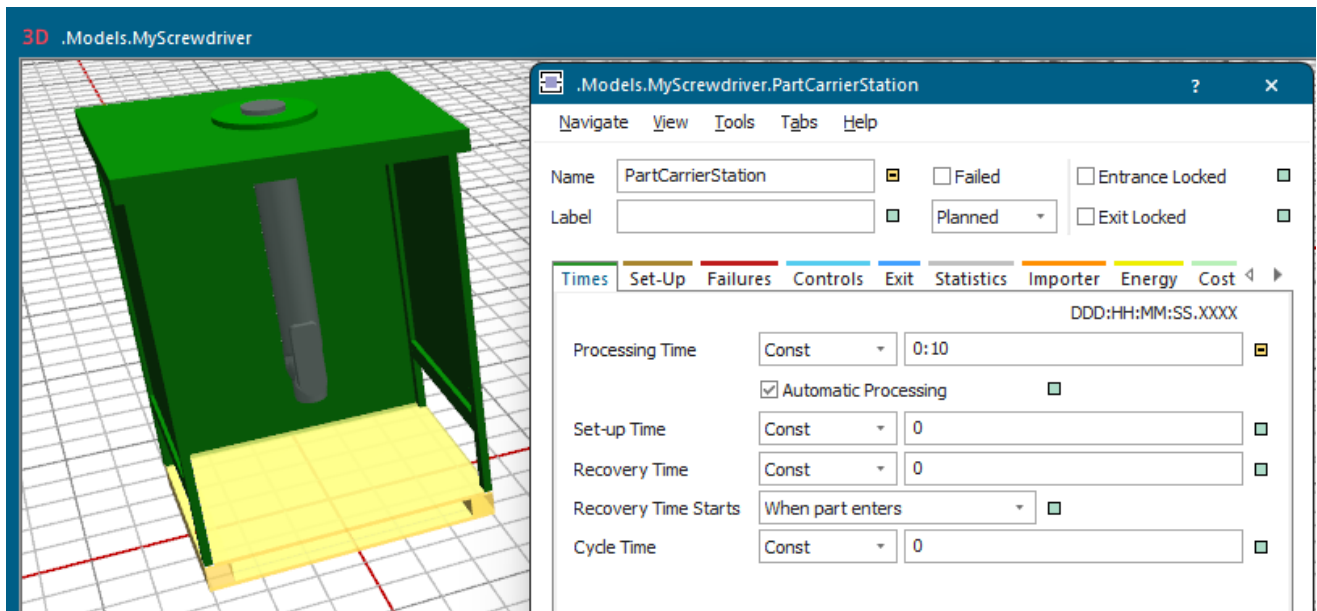
- Open the **Trestle** in a new window. Click the **Arm** with the right mouse button and change to the **Self Animation**.
  - **Add** four animation paths of type **Lines**. We named our paths **Setup**, **DownAdvance**, **Reset**, and **DownFinal**.
  - Select the respective path and click **Edit** to define two anchor points for each path. Click **Add** in the dialog **Path Anchor Points**, then select the new anchor point, and click **Edit Values**. Enter these values into the dialog that opens:
    - 0, 0, 0 to 0, 0, 0.9** for **Setup**.
    - 0, 0, 0.9 to 0, 0, -0.1** for **DownAdvance**.
    - 0, 0, -0.1 to 0, 0, -0.22** for **DownFinal**.
    - 0, 0, -0.22 to 0, 0, 0** for **Reset**.



With these settings, the animation paths of the arm resemble components of a continuous smooth up and down movement of the arm.



- Change to the **MU Animation** [tab], add a new path, name it Default, and add a single animation point with the settings 0, 0, -0.38.
- Finally, we are going to configure the simulation side of the stations, i. e. the stations we created as simulation objects above.  
To do so:
  - Double-click the **PartCarrierStation** and configure it. We entered a **Processing Time** of 10 seconds (0:10).



We also created the **Exit Control** called **self.OnExit** and entered this source code:

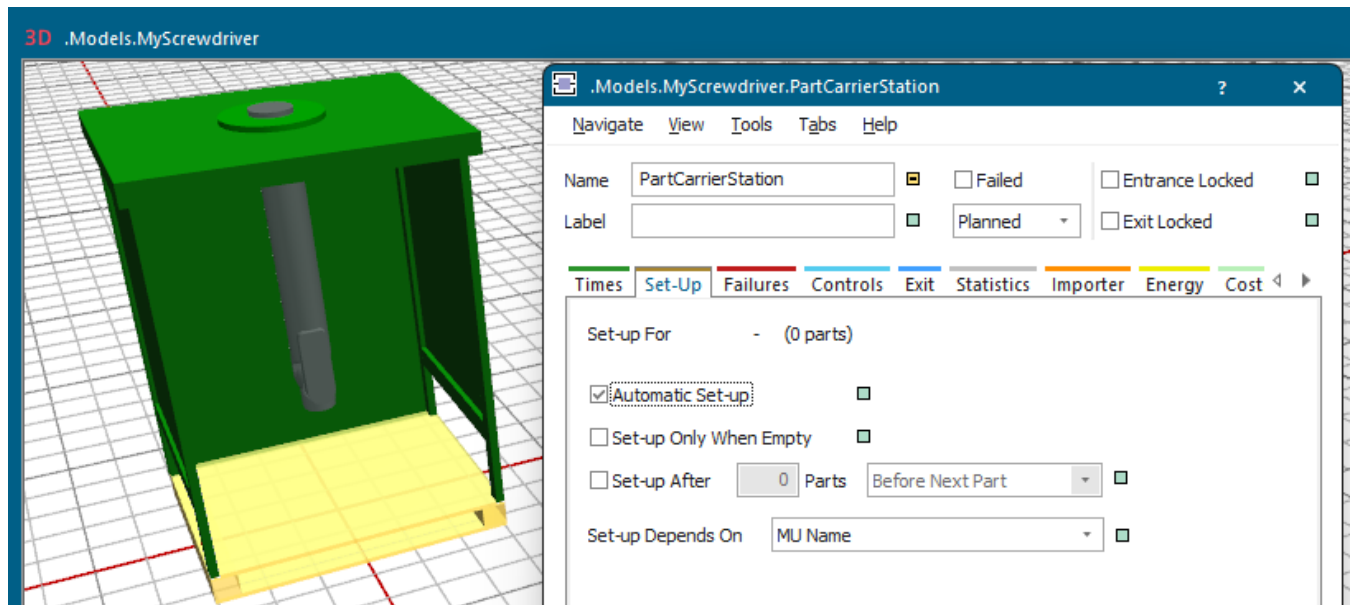
```
if ?.ExitCtrlFront
    self.~.Trestle.EntranceLocked := false
    @.move
end
```

In addition we added a *user-defined attribute* of data type *method*, named it **init**, and entered this source code:

```
self.~.EntranceLocked := true
```

- Double-click the **Trestle** and configure it. We entered a **Set-up Time** of 6 seconds (0:06) and selected these settings on the tab **Set-Up**.





- We also created the **Entrance Control** called **self.OnEntrance**, the **Exit Control** named **self.OnExit**, and the **Setup Control** called **self.OnSetup**.

We entered this source code into the **Entrance Control**:

```
var animations : any := self.~._3D.getObject("Arm").SelfAnimations
animations.resetAnimation
animations.DownAdvance.schedule
animations.startNextAnimationBlock
animations.DownFinal.schedule
animations.scheduleRotation(0, 360, 90)
@.outIn(animations.AnimationTimeTotal)
animations.playAnimation
```

We entered this source code into the **Exit Control**:

```
if ?.ExitCtrlFront
  @.deleteObject
  var animations : any := self.~._3D.getObject("Arm").SelfAnimations
  animations.resetAnimation
  animations.Reset.schedule
  animations.playAnimation
  self.~.EntranceLocked := true
end
```

We entered this source code into the **Setup Control**:

```
param setupStart: boolean
if setupStart
  var animations : any := self.~._3D.getObject("Arm").SelfAnimations
  animations.resetAnimation
```

```
    animations.Setup.schedule
    animations.playAnimation
else
    self.~.~.PartCarrierStation.EntranceLocked := false
end
```

In addition we added a *user-defined attribute* of data type *method*, named it `init`, and entered this source code:

```
self.~.EntranceLocked := false
```

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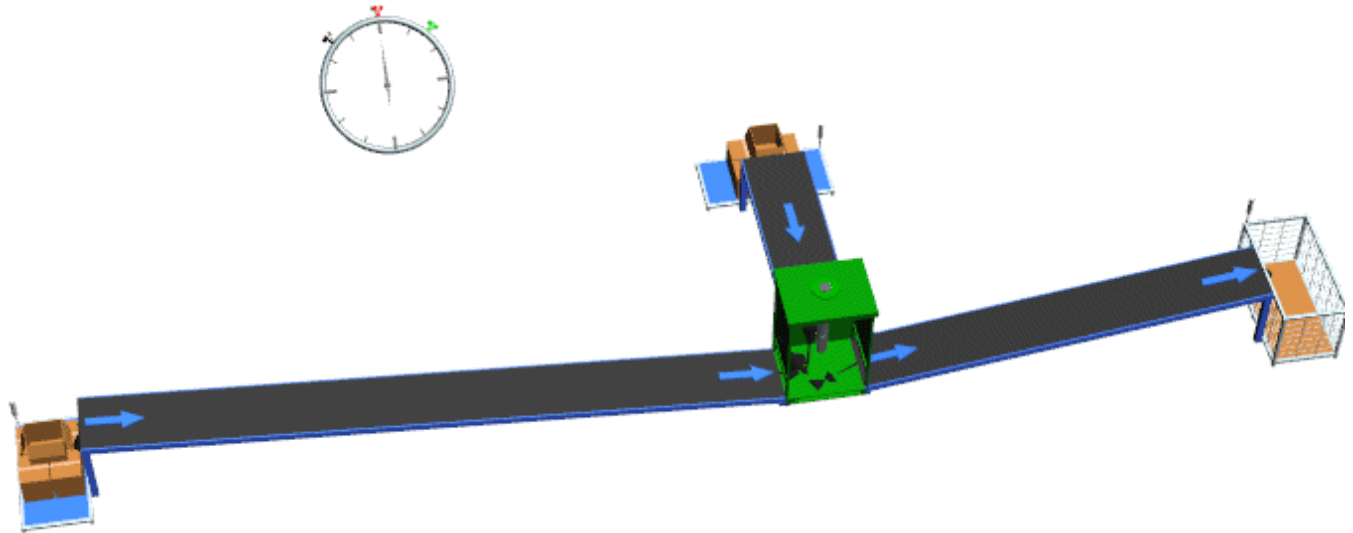
## Test the Station in the Simulation Model

Now we are ready to test our screwdriver station in the simulation model.

- Insert a *Source*, which produces the pallets which we want to screw together. We named it `SourcePallets`.

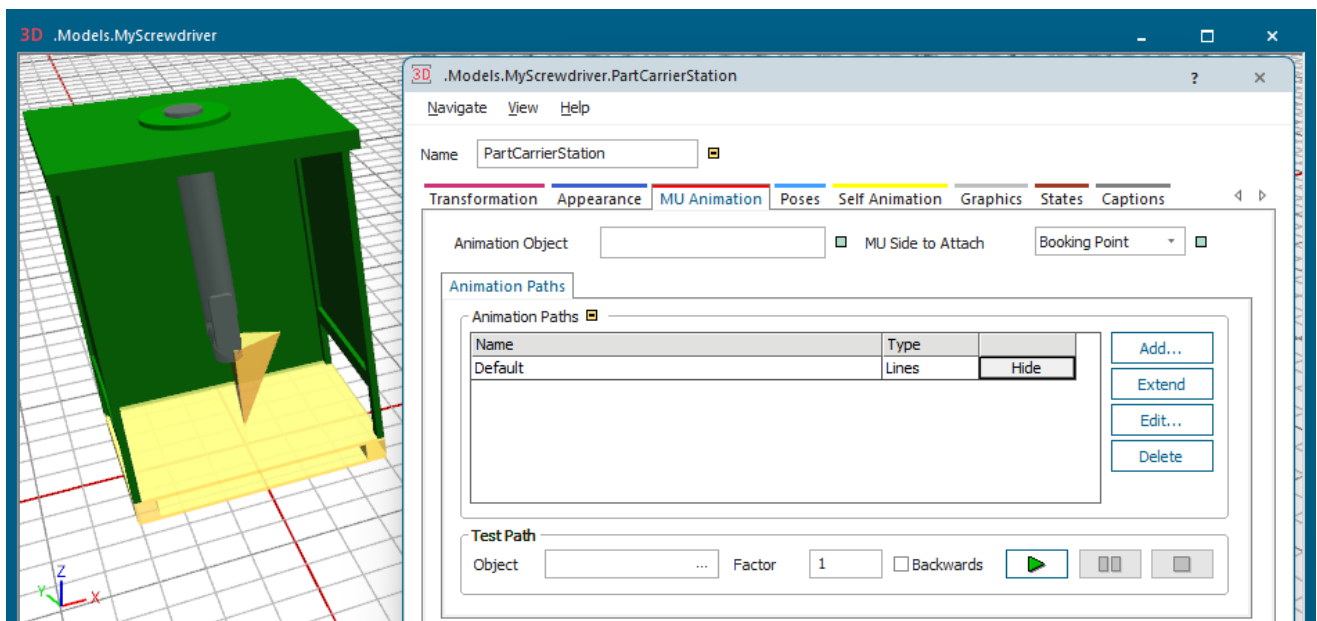
Insert a second *Source*, which feeds the screws required for screwing together the pallets. We named it `SourceScrews`. We received the screw, including the JT file, from a colleague.

- Insert a *Drain*, which removes the finished pallets from the plant.
- Insert our *Screwdriver* station.
- Insert three *Conveyors* between the *Sources*, the *Drain*, and the *Screwdriver* station.
- Connect all objects with *Connectors*. Connect the *SourcePallets*, that produces the **MU > PartsCarrier**, with the *ConveyorPallets*, and with the *InterfaceInPallet* of the *Screwdriver*. Connect the *SourceScrews* that produces the **MU > Screw**, with the *ConveyorScrews*, and with the *Interface* named *InScrew*. Connect the *ConveyorOut* with the *Interface* named *Out* and then with the *Drain*. The simulation model should now look like this:



You might be wondering how we inserted the *Conveyors* with an inclination to smoothly connect to the *Screwdriver* station. We proceeded as follows:

- For the *ConveyorPallets* we entered an **Anchor Point Height** of 1 meter for the first anchor point and an anchor point height of 0.15 meters for the second anchor point into the dialog **Line/Arc Parameters**. This is the height of the object *SourcePallets* to which we connect it on the left-hand side.
- For the *ConveyorOut* we edited values in the dialog **Edit 3D Properties > Appearance**. We entered a **Base Height** of 0.15 meters, which means that the left-hand side of the *ConveyorOut* is placed 15 centimeters above the floor. We then clicked **Segments** and entered 1 meter as the **Z** dimension so that the right-hand side connects smoothly to the *Drain*.



- When you run the simulation, you might want to run it with a low **real-time** factor to be able to watch how the screwdriver actually screws the screws in.

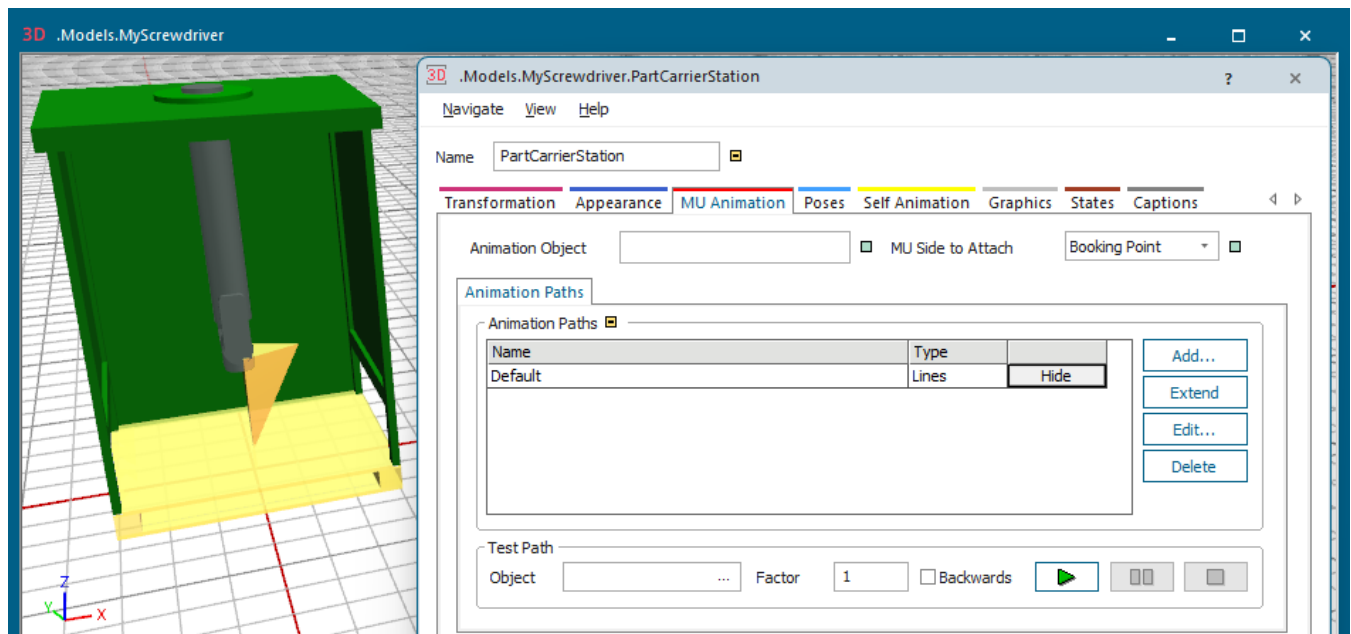
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### Fine-tune the Station

While running the simulation, you'll notice that the pallet is not located at the bottom of the *Screwdriver* station, but at the top.

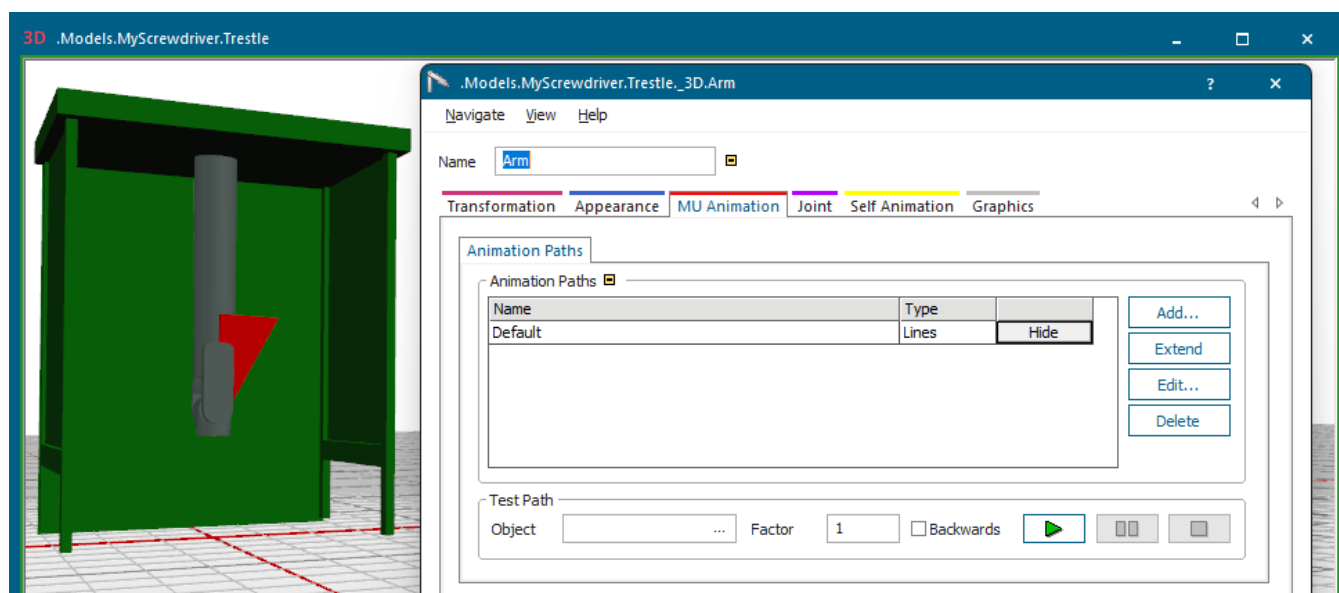
Proceed as follows to fix this:

- Open the class of the *Screwdriver* in a new window.
- Then, open the dialog **Edit 3D Properties** of the **PartCarrierStation**, change to the tab **MU Animation**, and **show** the **Default** path.
- Select the path marker, i. e., the wedge pointing downwards, hold down the **Ctrl** key, and move it down with the down arrow key until it is located flush on the **PartCarrierStation**.



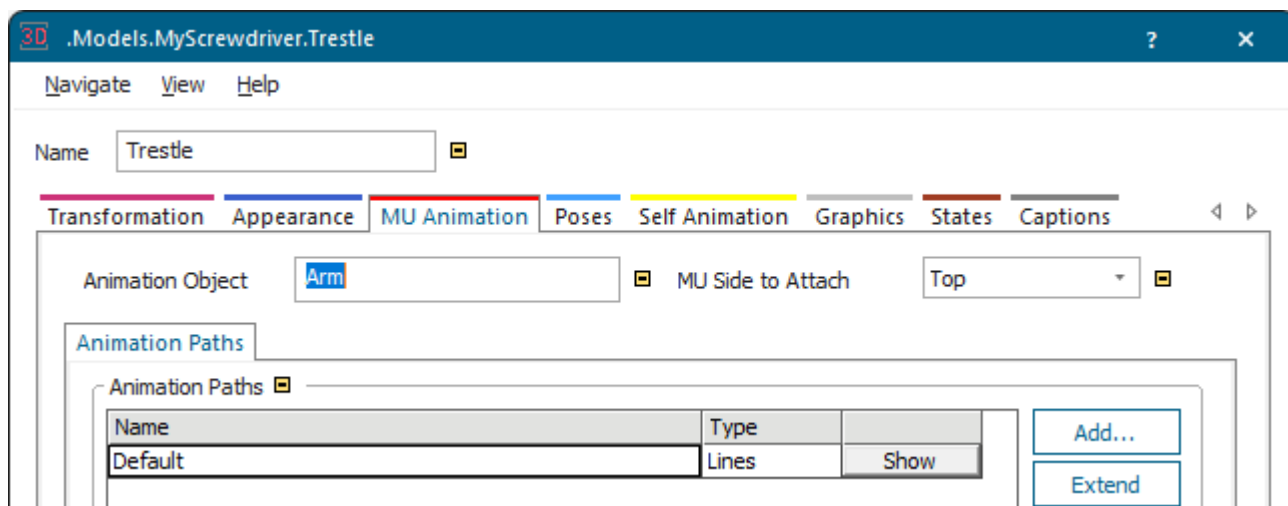
You'll also notice that the screw is not attached to the tip of the arm. Proceed as follows to fix this:

- Open the class of the *Screwdriver* in a new window.
- Open the **Trestle** of the *Screwdriver* in a new window.
- Open the dialog **Edit 3D Properties** of the **Arm**, change to the tab **MU Animation**, and visualize the **Default** path.
- Select the path marker, hold down the **Ctrl** key, and move it down with the down arrow key until it was located flush at the tip of the **Arm**.

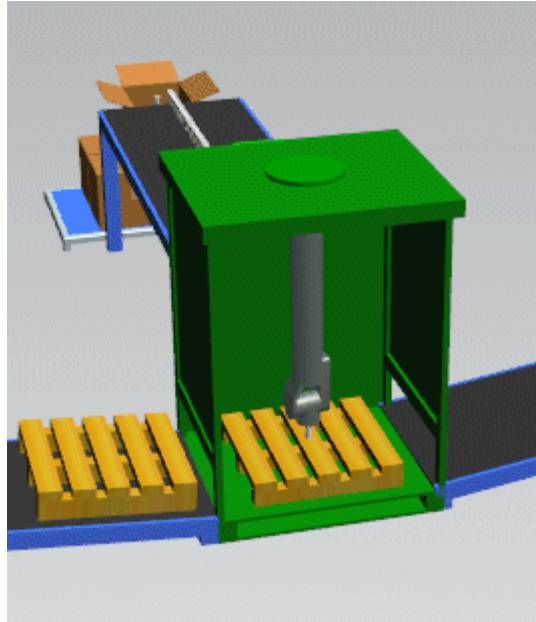


You'll finally notice that the wrong side of the screw is attached to the tip of the *Screwdriver* arm. Proceed as follows to fix this:

- Open the class of the *Screwdriver* in a new window.
- Open the **Trestle** of the *Screwdriver* in a new window.
- Open the dialog **Edit 3D Properties** of the **Trestle**, change to the tab **MU Animation**, and **visualize** the **Default** path.
- Select **Top** as the **side** to which the **MU** is to be **attached**.



Run the simulation again. Now our *Screwdriver* station does exactly what we designed it to do.



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## Modeling a Complex Receiving Department

### Model a Complex Receiving Department

This sample model demonstrates how to model a complex receiving department in a *Frame*.

Here three *Sources* produce three different *parts*, namely *bricks*, *glass bricks*, and *brick carriers*. The *bricks* and the *glass bricks* are then loaded onto *brick carriers* and moved to a *ParallelStation* that stores them before a *TransferStation* loads them onto a *truck* that is produced by a fourth *Source*.

We will demonstrate how to:

- [Insert the Objects Required for Modeling the Source](#)
- [Configure the Individual Stations](#)
- [Animate Bricks on the Animation Area of the Brick Carrier](#)
- [Animate Brick Carriers on the Animation Area of the Truck](#)
- [Show the Content of the Frame](#)

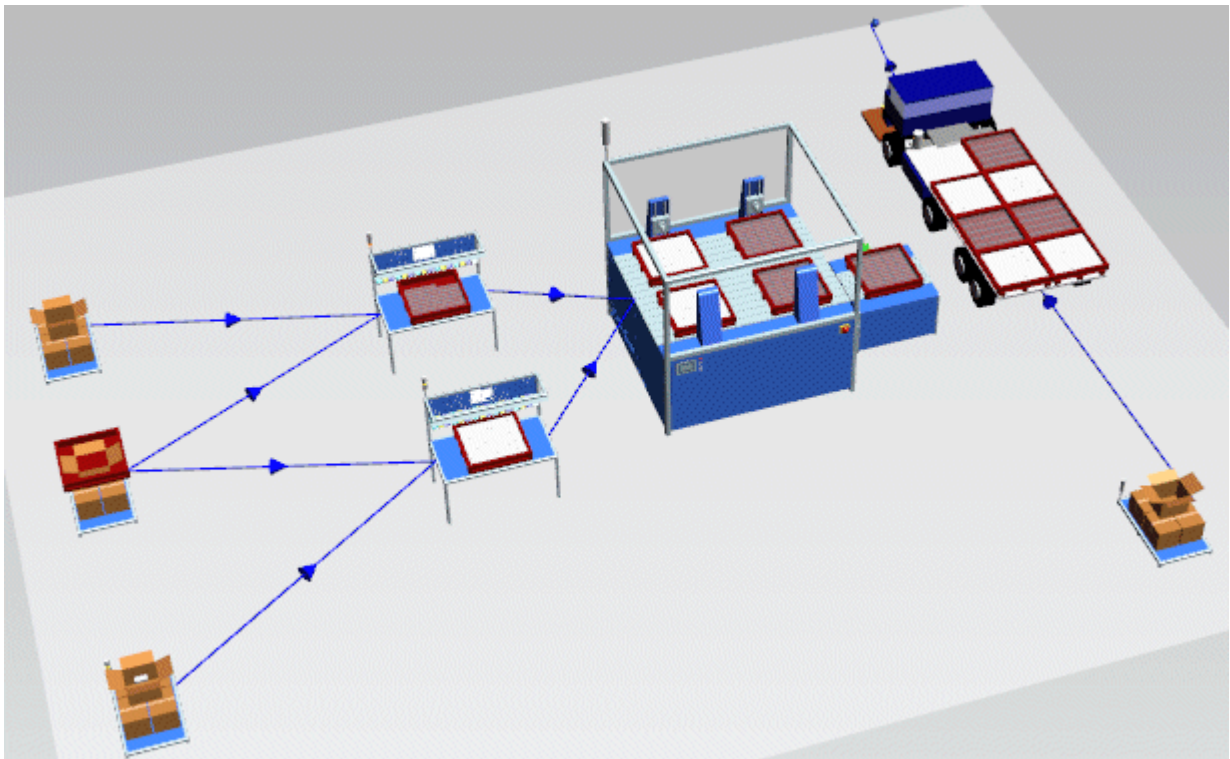
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### Insert the Objects Required for Modeling the Source

First, we will insert the objects, which we need to model our receiving department, into a *Frame*.

In the second step, we are going to configure the individual objects.

- Create a new *Frame* in the *Class Library* of your simulation model. We named our *Frame* *MySourceTruckLoadedWithBricks*.
- Insert three *Sources* at the left-hand side of the *Frame*. We use our own class, named *MySource*, with a modified 3D graphic. We named our *Sources* *SourceGlassBricks*, *SourceBricksCarrier*, and *SourceBricks*. They will produce glass bricks, brick carriers, and normal bricks.
- Insert two *AssemblyStations* to the right of the *Sources*. We named them *AssemblyGlassBricks* and *AssemblyBricks*. Each of them places **50 glass bricks** or **50 normal bricks** onto a *pallet* each.
- Insert a *TransferStation*, a short *Track*, and another *Source*. This *Source*, which we named *SourceTrucks*, produces the *trucks* onto which the commissioned pallets are loaded.
- Insert an *Interface* that connects our *receiving department* with the other objects in the simulation model.
- Connect the objects so that they match the screenshot below. Our finished model looks like this:



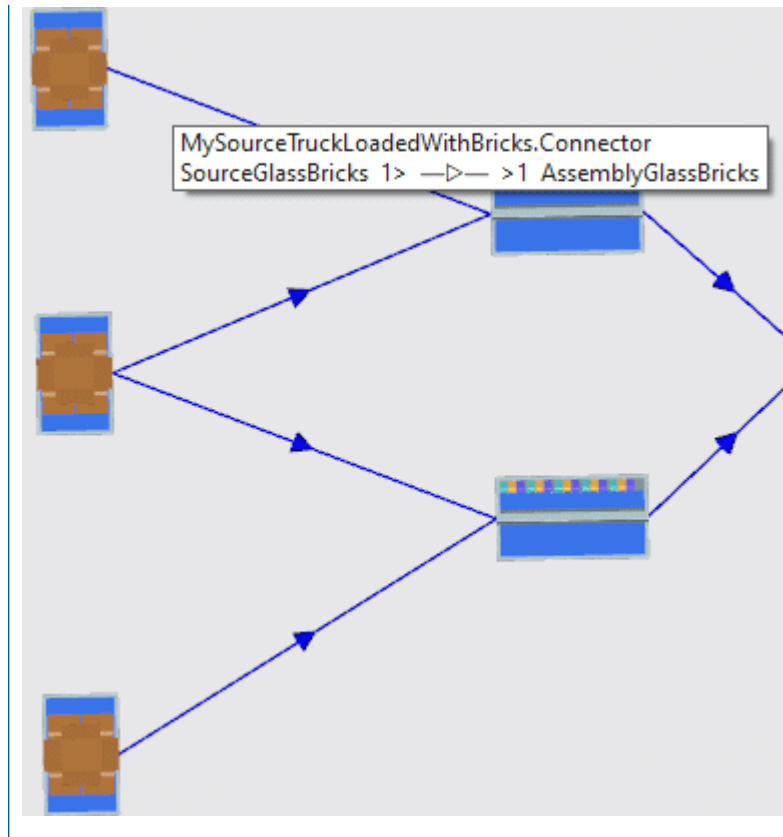
#### Note

As the *AssemblyStation* is connected to two stations, the order in which you connect it with the predecessors is important. If the assembly process does not work as expected, check the numbering of the predecessors.

*Plant Simulation* shows the **predecessors** and the **successors** of an object in a *tooltip* when you drag the mouse over the respective *Connector*.

Then, enter the correct number of the predecessor into the text box **Main MU from Predecessor** of the *AssemblyStation*.





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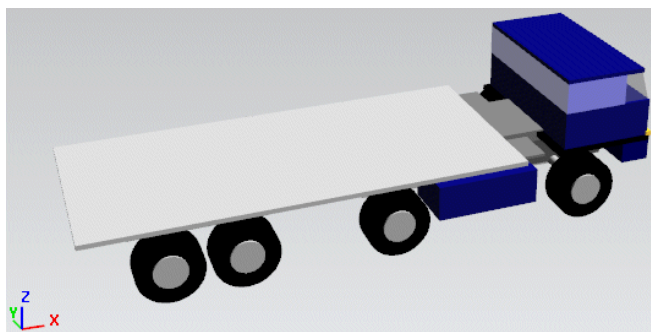
## Configure the Individual Stations

After we have inserted the objects we need to model our receiving department, we are going to configure the individual stations.

First, we are going to configure the *MUs*:

- To create the *MU* representing the *truck*, duplicate the class **.MUs.Transporter** and rename it to **Truck**. We deleted its graphic and imported the JT file *Truck.jt*, which we received from a colleague. Position the graphic in its window as shown below. We changed its size to 6.3 by 2.2 by 1.6 meters.





**.MUs.Truck**

Navigate View Tools Tabs Help

Name  ☐ Stopped  
 Label  ☐ Failed

Attributes

MU Size

Length  m  
 Width  m  
 Height  m

Booking Point

[0..1] (3.15 m)  
 [0..1] (1.10 m)  
 [0..1] (0.00 m)

Speed  m/s ☐ Backwards  
☒ Acceleration  m/s<sup>2</sup>  
 Current Speed  m/s Deceleration  m/s<sup>2</sup>  
☐ Is Tractor  
 Start Delay Duration

3D OK Cancel Apply

- To create the *MU* representing the *brick carrier*, duplicate the class **.MUs.Container** and rename it to **BricksCarrier**. We deleted its graphic and imported the JT file *BricksCarrier.jt*. We changed its size to 1.1 by 1.1 by 0.15 meters and gave it a capacity of 50 parts (X-Dimension of 5 times Y-Dimension of 10).

**.MUs.BricksCarrier**

Navigate View Tools Help

Name  ☐ Stopped  
 Label  Conveying Direction

Attributes

MU Size

Length  m  
 Width  m  
 Height  m

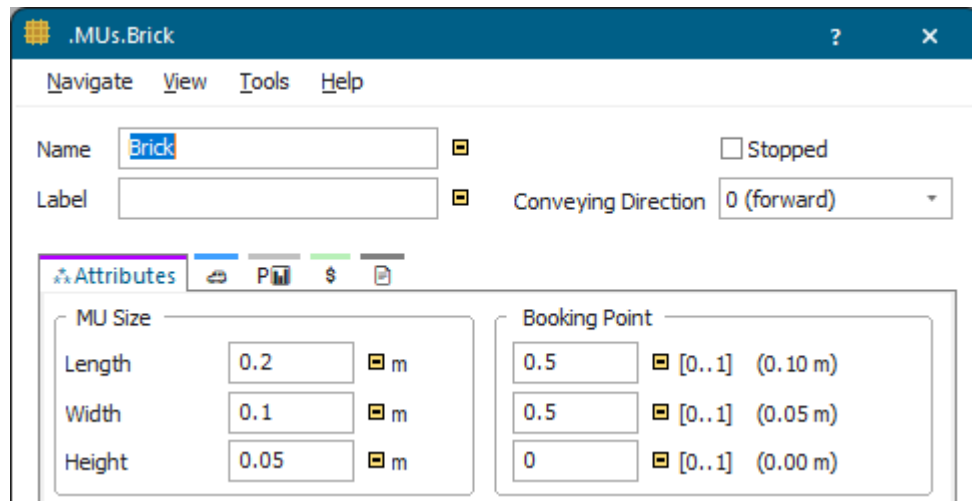
Booking Point

[0..1] (0.55 m)  
 [0..1] (0.55 m)  
 [0..1] (0.00 m)

X-Dimension   
 Y-Dimension   
 Z-Dimension

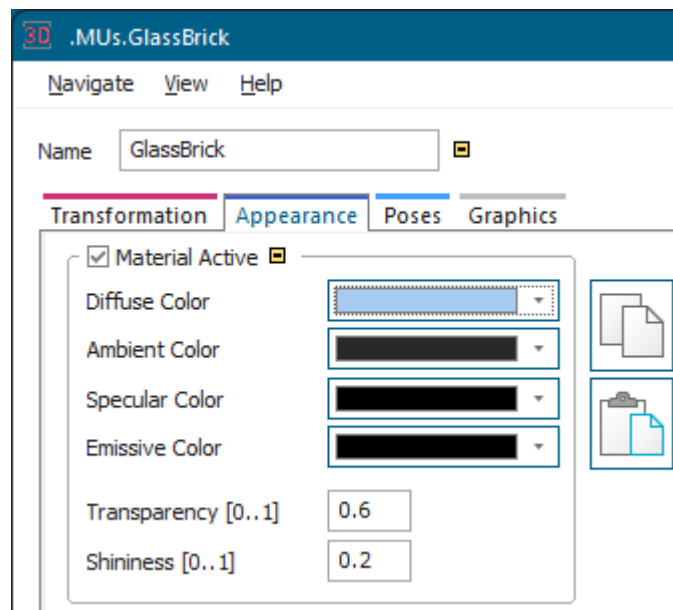
☐ Workpiece Carrier  
☐ Fill Whole Layer

- To create the *MU* representing the *normal brick* and the *glass brick*, duplicate the class **.MUs.Part** twice. We renamed them to **Brick** and **GlassBrick** respectively. Both have a size of 0.2 by 0.1 by 0.05 meters.



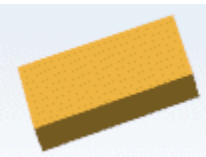
For the bricks we are going to use automatically generated graphics instead of importing JT files.

- Open their class in a new window and open the dialog **Edit 3D Properties**.
- Change to the tab **Transformation** and activate the check box **Scale Automatically**.
- For the *GlassBrick*, also activate **Material Active**, select a blueish gray as the **Diffuse Color**, and select a **Transparency** factor of 0.6 to make the glass brick semi-transparent.



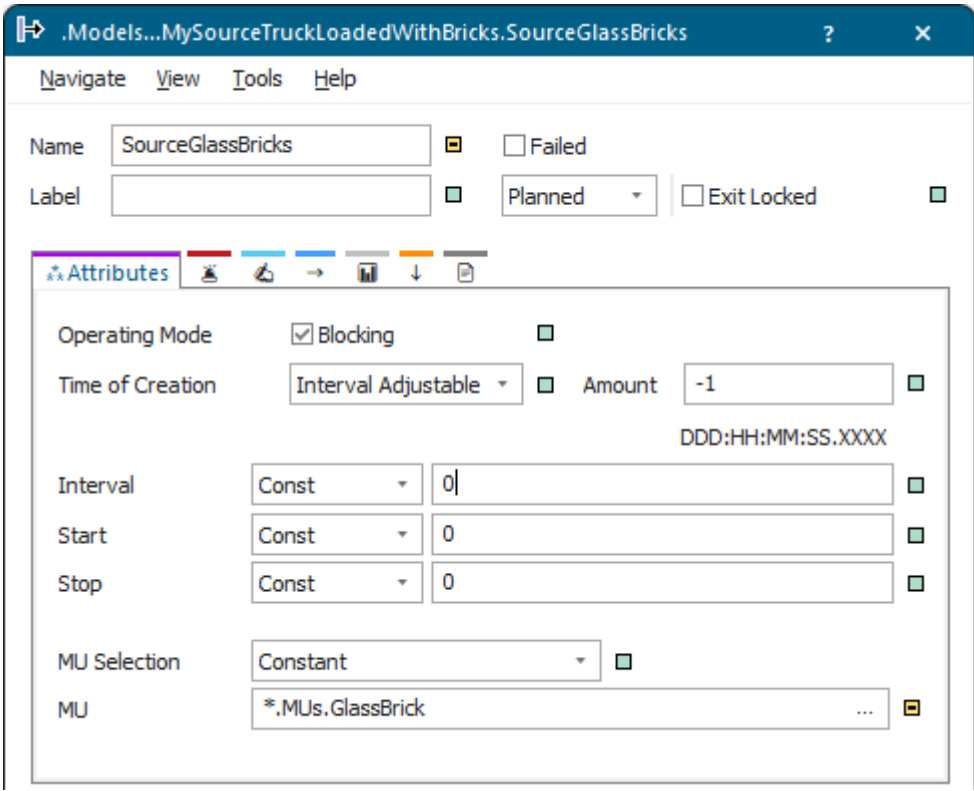
- Activate **Material Active** for the *GlassBrick*. Select orange as the **Diffuse Color**

The automatically created graphics look like this:

| Brick                                                                             | Glass brick                                                                        |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |

Then, we are going to configure the other objects in our *Receiving department*.

- Start with the *Sources* producing the *Brick*, the *GlassBrick*, and the *BricksCarrier*. Select the respective **MU** according to the name of the *Source*. We did not change any other settings. Compare the example below.



Attributes

Name: SourceGlassBricks ☐ Failed

Label:  ☒ Planned ☐ Exit Locked

Operating Mode ☐ Blocking ☐

Time of Creation: Interval Adjustable ☐ Amount: -1 ☐

Interval: Const  ☐

Start: Const  ☐

Stop: Const  ☐

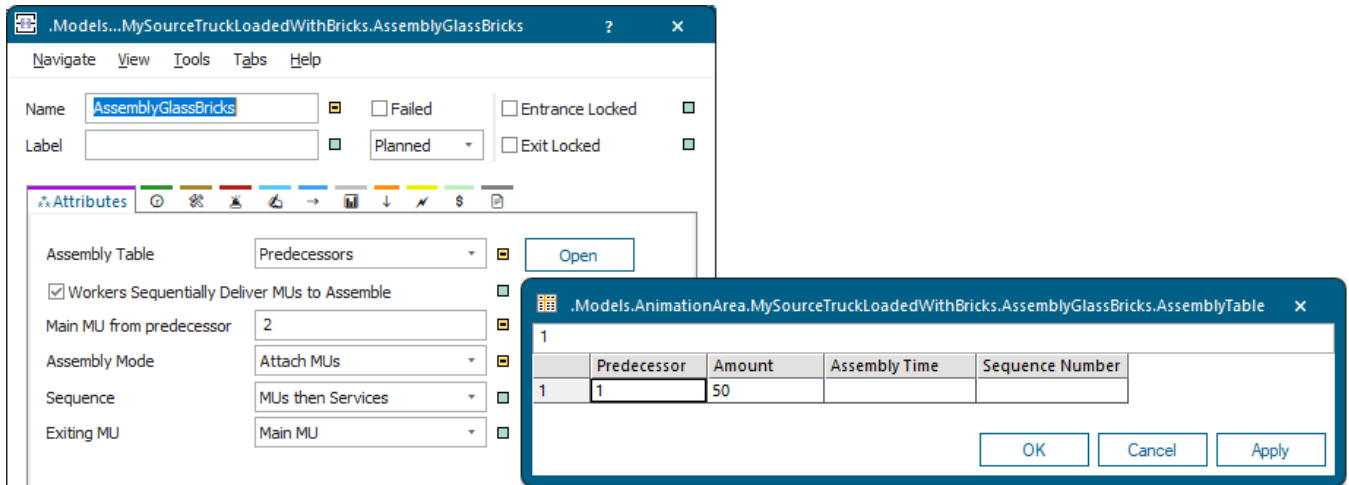
MU Selection: Constant ☐

MU: \*.MUs.GlassBrick ☐

#### Note

Make sure that you use the **absolute path** for the *MUs*, denoted by an asterisk \* at the beginning of the path. The relative path starts with a tilde ~.

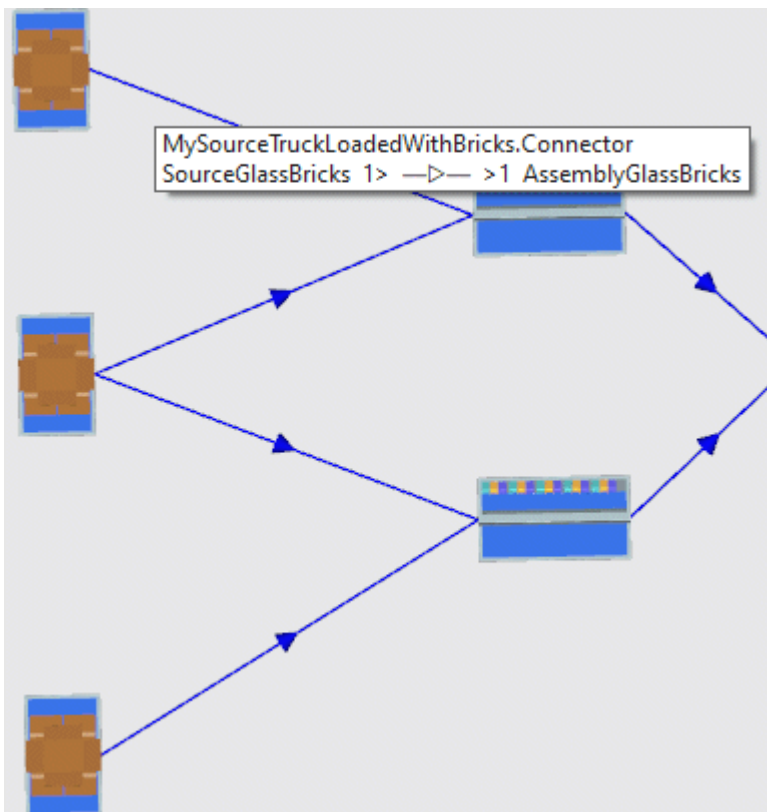
- Configure the *AssemblyStations* to **attach the MUs** and to make the **Main MU** exit the station. We also want them to load 50 parts, namely the *bricks* or *glass bricks*, onto the *brick carrier*. Click **Open** next to **Assembly Table** and enter 50 into the column **Number**.



### Note

As the *AssemblyStation* is connected to two stations, the order in which you connect it with the predecessors is important. If the assembly process does not work as expected, check the numbering of the predecessors.

*Plant Simulation* shows the **predecessors** and the **successors** of an object in a *tooltip* when you drag the mouse over the respective *Connector*.



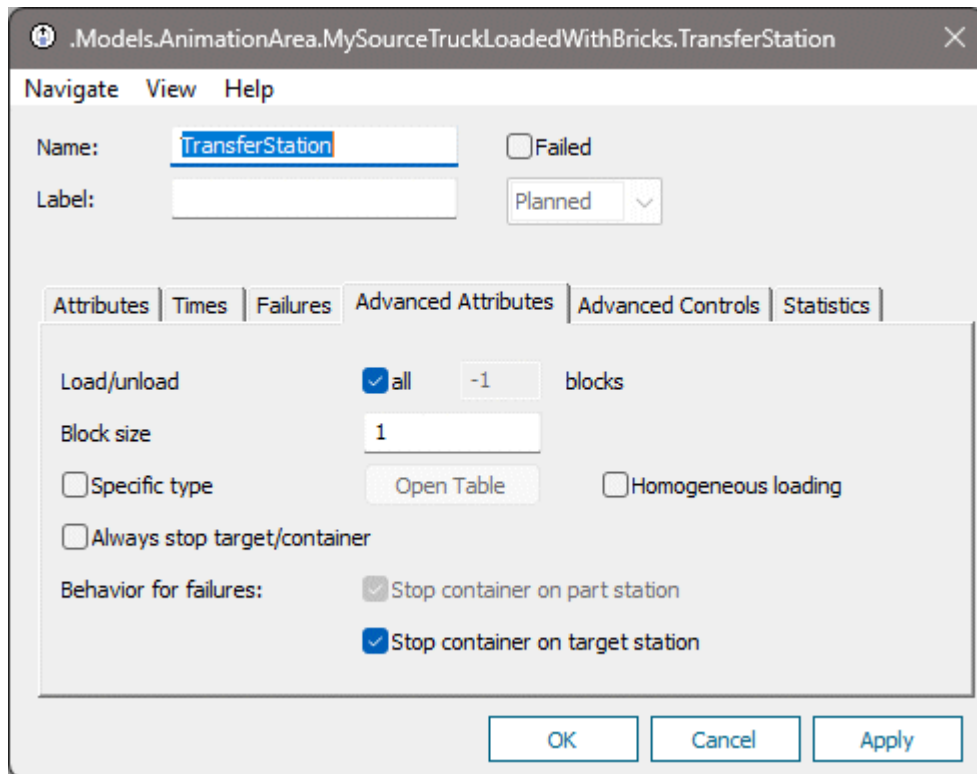
Then, enter the correct the number of the predecessor into the text box **Main MU from Predecessor** of the *AssemblyStation*.

- Drag the *ParallelStation* onto the *TransferStation* to **load** parts and to use it as the **Part Source**. We use the default settings of the *ParallelStation*.

Drag the *Track* onto the *TransferStation* to use it as the **Target Station**. Accept 0 is the **Sensor Position**.

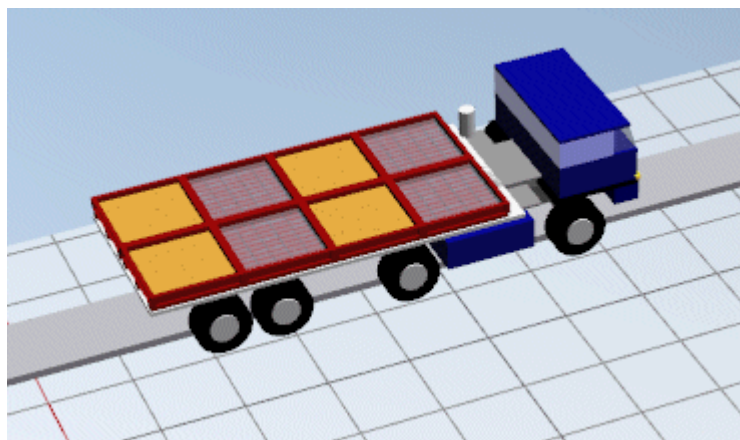
The screenshot shows the configuration window for a TransferStation. The title bar reads ".Models.AnimationArea.MySourceTruckLoadedWithBricks.TransferStation". The window has a menu bar with "Navigate", "View", and "Help". Below the menu bar, there are fields for "Name:" (TransferStation) and "Label:" (empty), along with a "Failed" checkbox and a "Planned" dropdown menu. A tabbed interface is present with tabs for "Attributes", "Times", "Failures", "Advanced Attributes", "Advanced Controls", and "Statistics". The "Attributes" tab is active, showing a "Station type" section with four radio buttons: "Load" (selected), "Unload", "Reload", and "Move". Below this, there are two sections: "Parts from:" and "Target is on:". "Parts from:" has a text field containing "ParallelStation" and a button with three dots. "Target is on:" has a text field containing "Track" and a button with three dots. At the bottom of these sections, there are "Sensor position:" labels followed by a text field and a dropdown menu set to "Lane A". The "Sensor position" text field for "Parts from:" is empty, while for "Target is on:" it contains the value "0". At the bottom right of the window are three buttons: "OK", "Cancel", and "Apply".

To make the *Truck* stop and wait for parts until it is fully loaded, select the check box **Always Stop Container** on the tab **Advanced Attributes**.



- Open the dialog **Edit 3D Properties** of the *Frame MySourceTruckLoadedWithBricks*. Change to the tab **Graphics** and make sure that **Show Content** is cleared.

Now we are ready to test our receiving station *MySourceTruckLoadedWithBricks*. Insert it into our simulation model, connect it with a *Track* and a *Drain*, and run the simulation. You'll notice that it creates *trucks* that transport two types of *carriers*, one loaded with *normal bricks* and one loaded with *glass bricks* each.

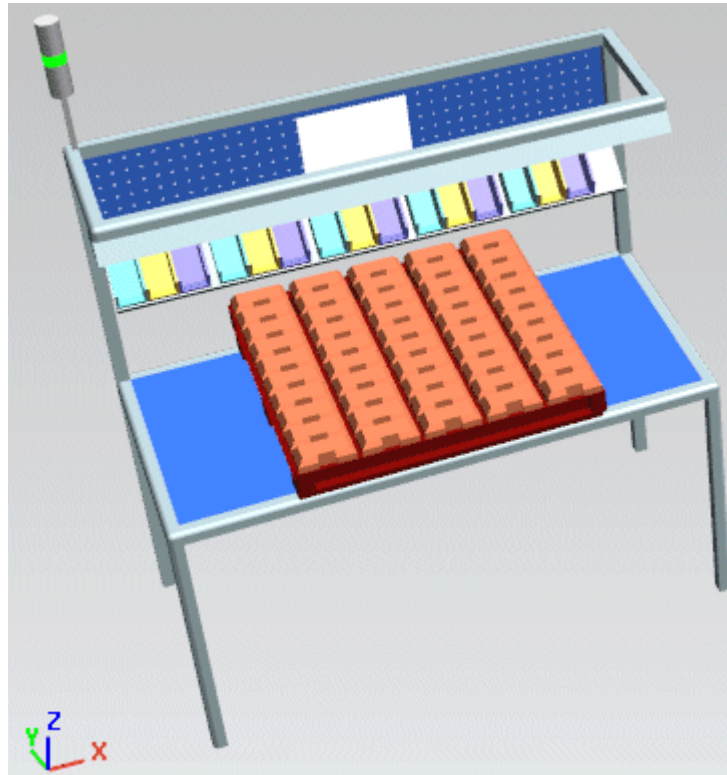


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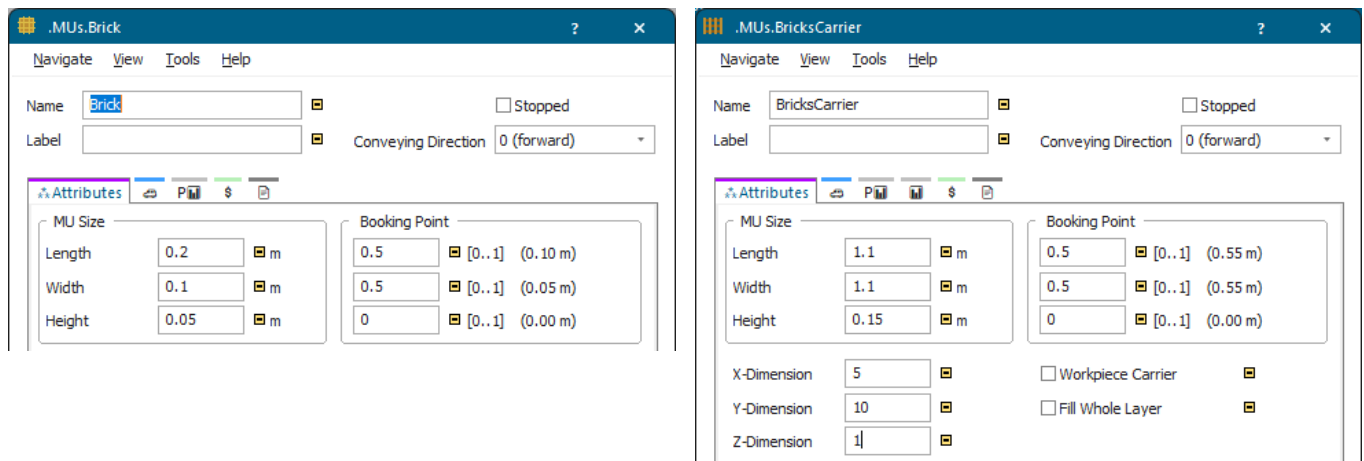
## Animate Bricks on the Animation Area of the Brick Carrier

When *Plant Simulation* places objects onto other objects, in our example *bricks* and *glass bricks* on the *brick carrier*, it automatically distributes them evenly across the **animation area** of the *brick carrier*.

*Plant Simulation* distributes them according to their **size** and to the **size** and **dimension** of the *brick carrier*.



In our example the *brick carrier* can accommodate 50 *bricks* on its **animation area**, each of which is 20 centimeters long and 10 centimeters wide. The brick carrier is 110 centimeters long and wide. According to its x-dimension 5 *bricks* fit in a row side by side in the x-direction. According to its y-dimension 10 *bricks* fit in a row in the y-direction.



We used the default settings of the *brick carrier/container*.

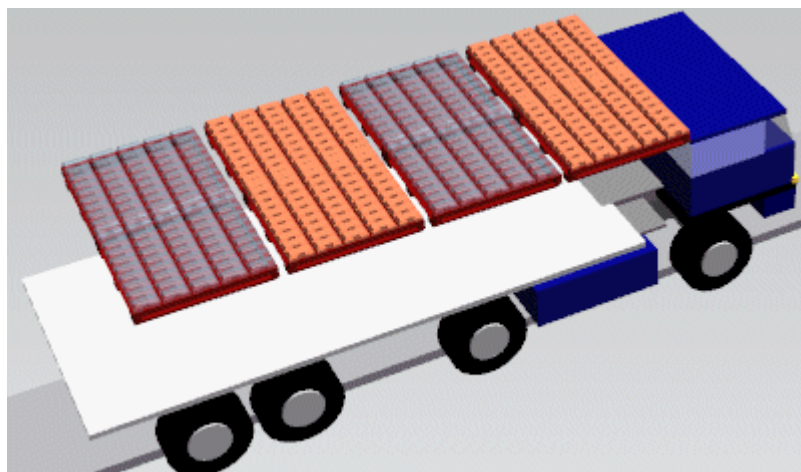
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## Animate Brick Carriers on the Animation Area of the Truck

When *Plant Simulation* places objects onto other objects, in our case *brick carriers* onto the loading space, i.e., the animation area of the truck, we would like to determine where on the bed it places the *brick carriers*.

We defined animation areas for the objects with matrix loading space *ParallelStation*, *Sorter*, *Store*, *Transporter*, and *Container*. These should cover your basic modeling needs.

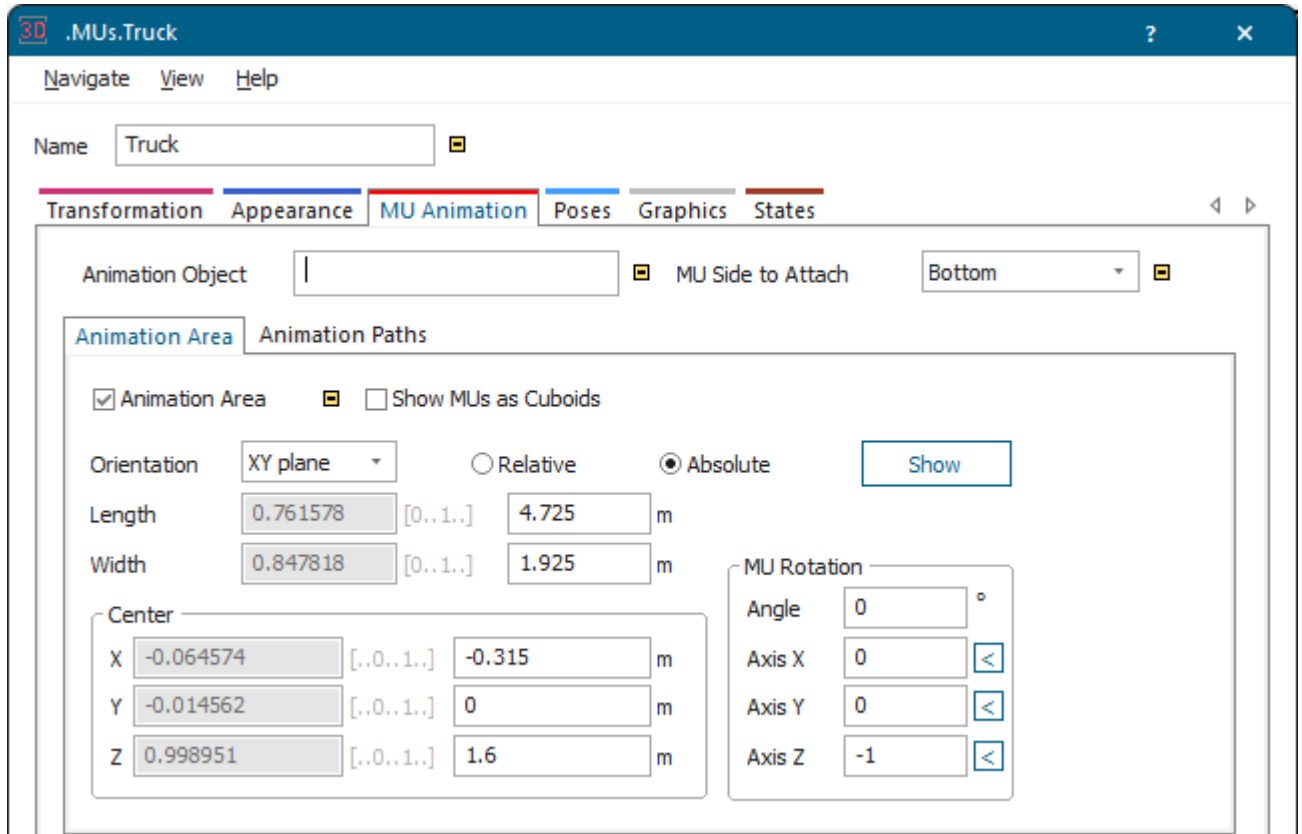
In our case we changed the graphic of the truck, which we derived from the *Transporter*. We thus have to change the animation area settings.



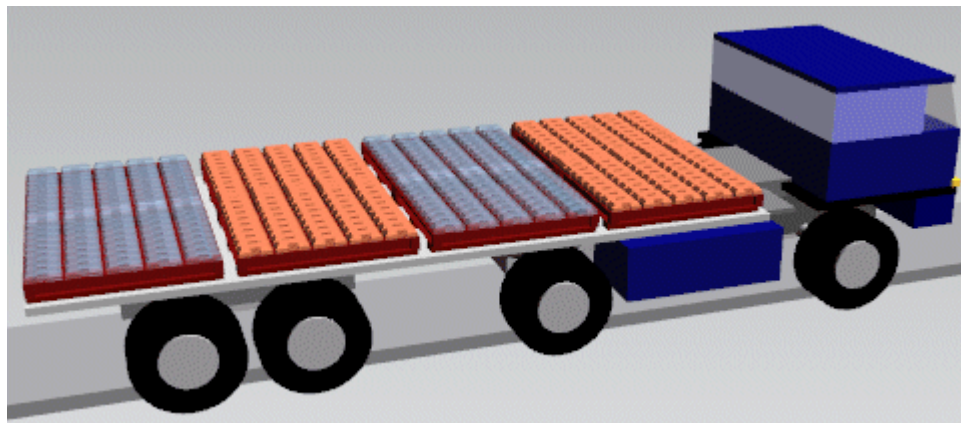
Proceed as follows:



- Change to the *Class Library* and click **MUs > Truck** with the right mouse button.
- Click **Open In 3D**.
- Click the right mouse button in the window **.MUs.Truck** and select **Edit 3D Properties > MU Animation**.
- You can now adjust the area settings under **animation area** so that the *brick carriers* are evenly distributed across the loading space of the truck. *Plant Simulation* computes the animation area of the object as soon as you click **Apply** in the dialog **Edit 3D Properties**. The window of our partial model then also shows these changes.



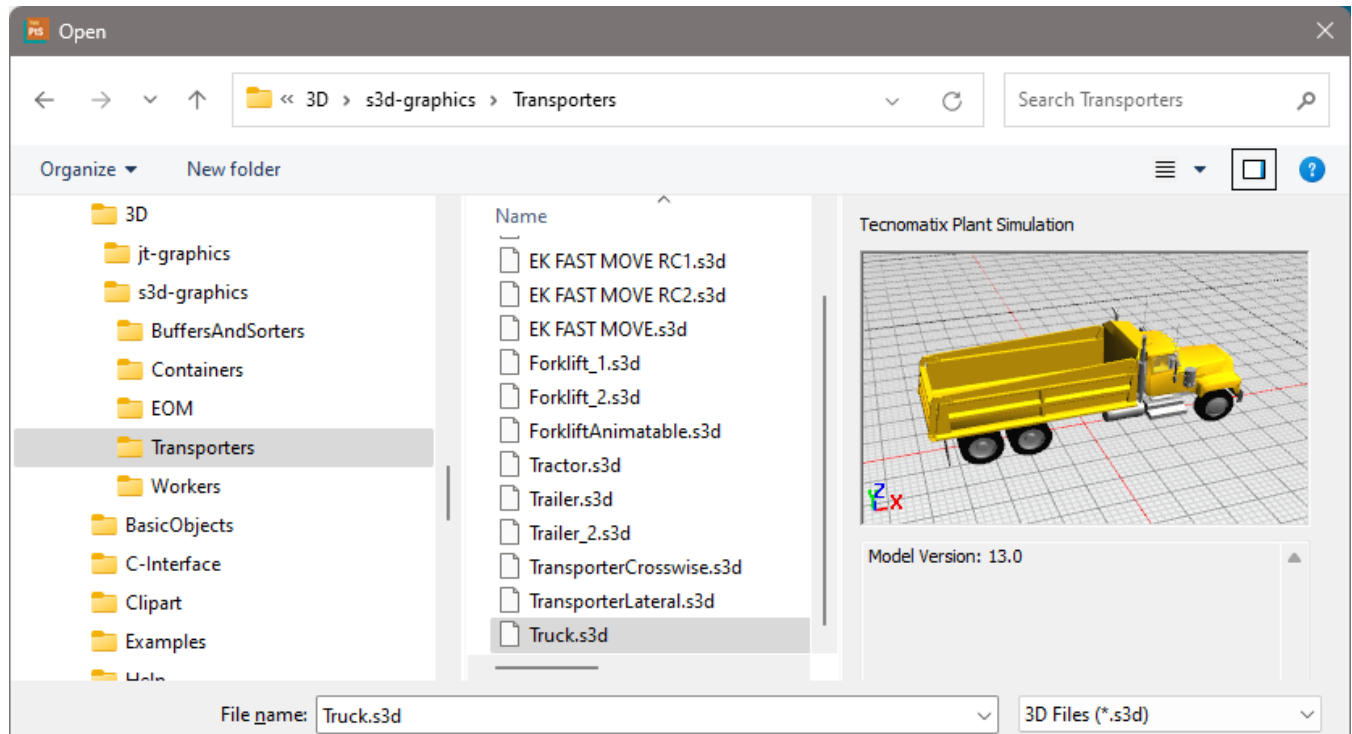
- We selected the settings above. Now everything is correct.



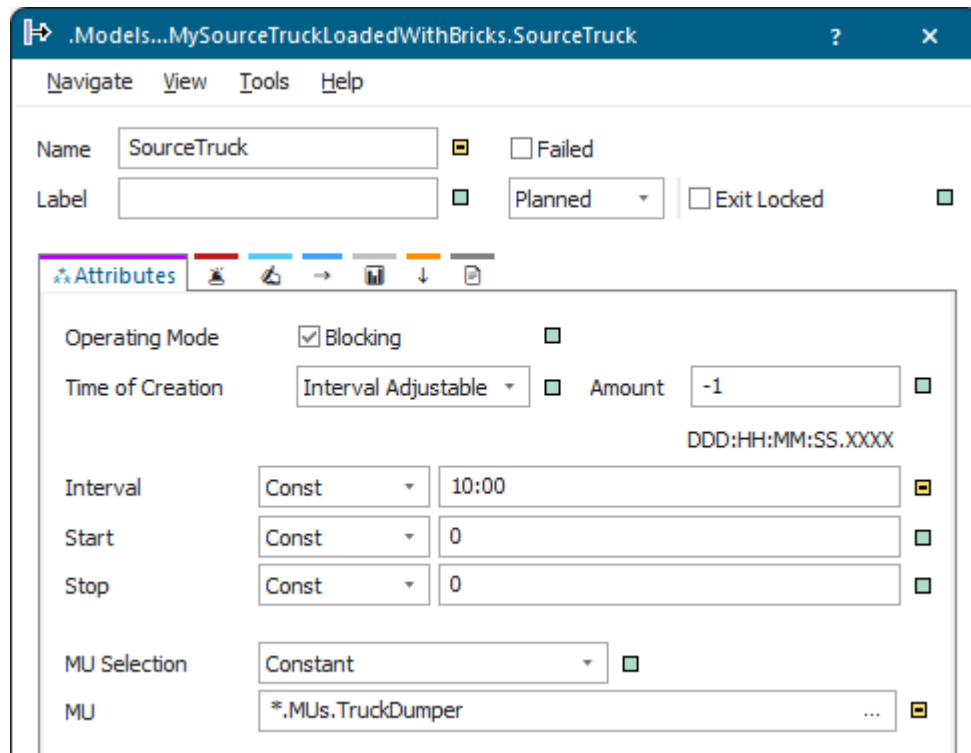
We can also replace the simple graphic of the truck with the graphic of the new version of *Plant Simulation*.

To do so:

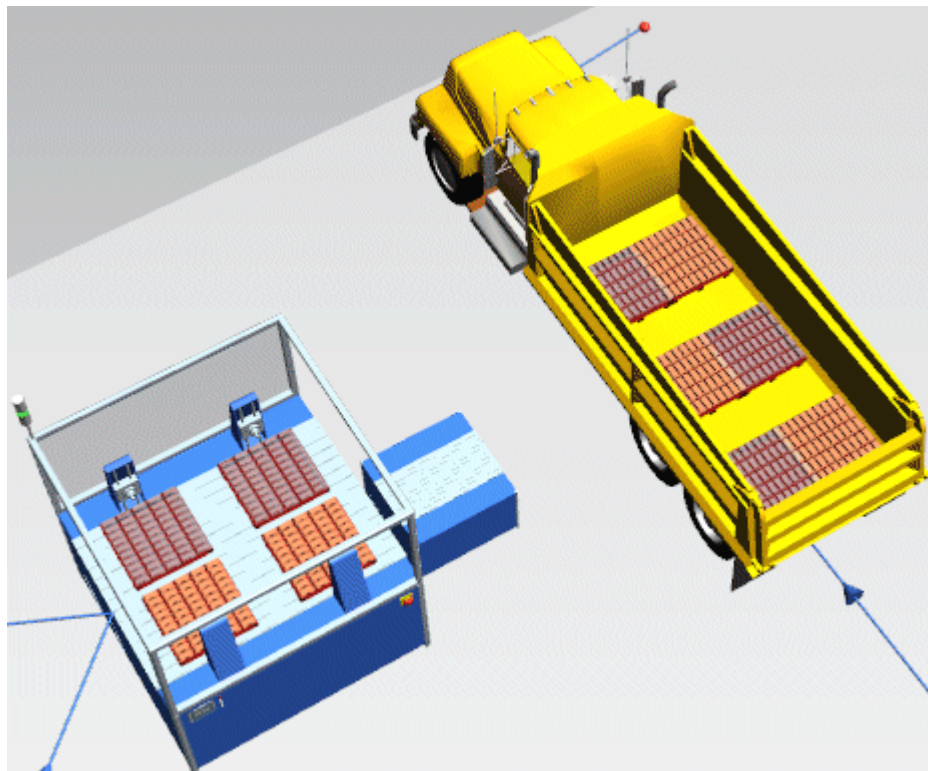
- Duplicate the *Transporter*, rename it to *TruckDumper*, and **exchange its graphic**.



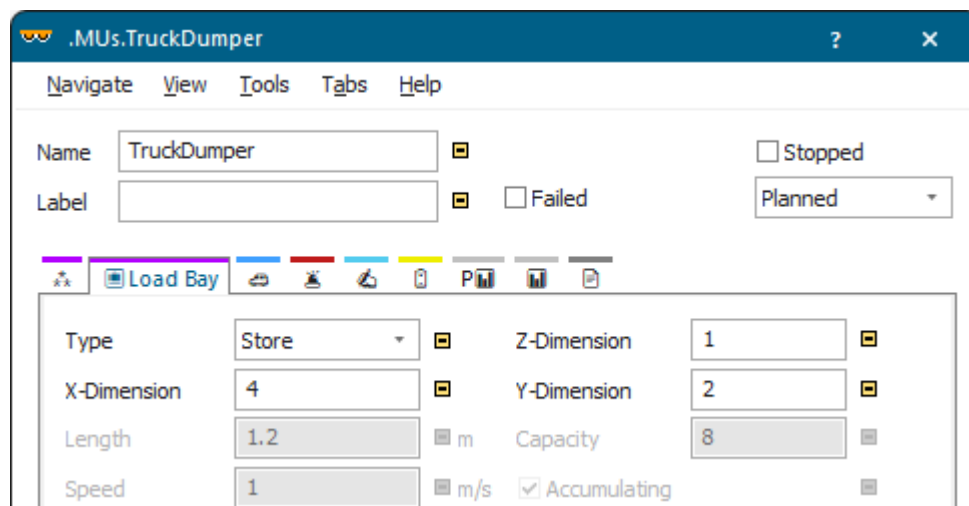
- Enter the dumper as the **MU** to be created into the *SourceTruck*.



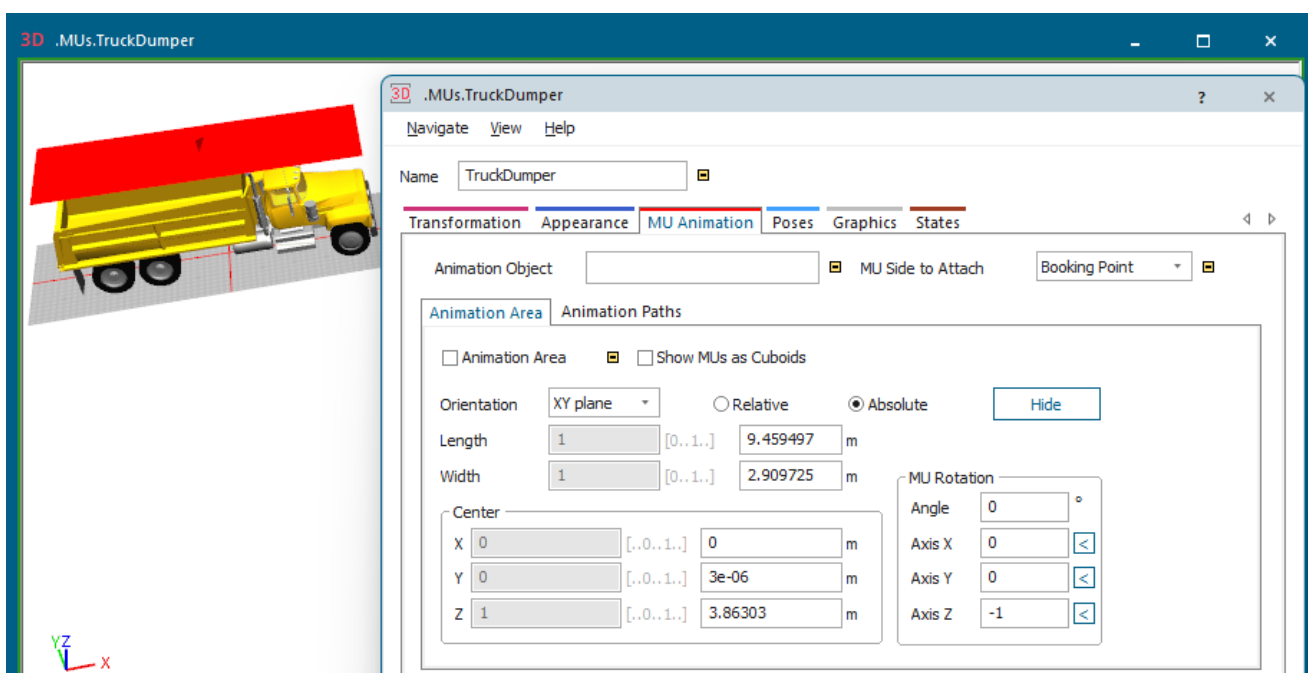
- Now only 6 brick carriers are loaded instead of the intended 8.



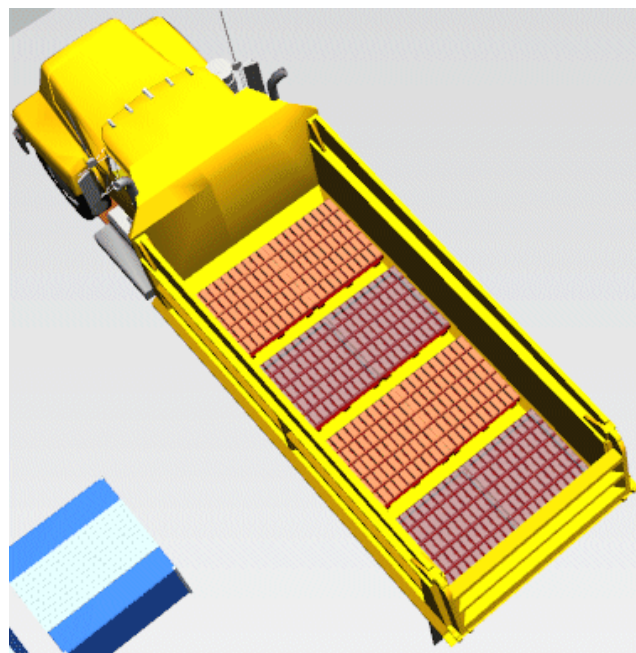
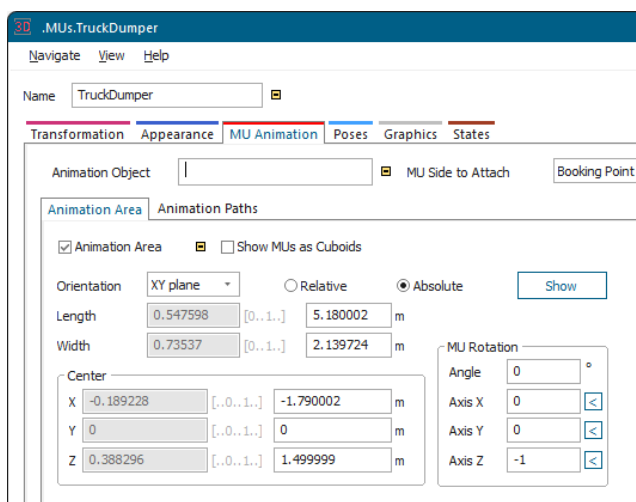
To fix this, increase the **X-Dimension** of the loading space from **3** to **4**.



- We finally adjust the animation area settings of the bed.
  - Open the *TruckDumper* in a new window, click in the background, and press the **spacebar**.
  - Activate the **Animation Area [described]** on the tab **MU Animation** and **Show** it.



- Click in the text boxes and roll the mouse wheel until the animation area has the correct size.



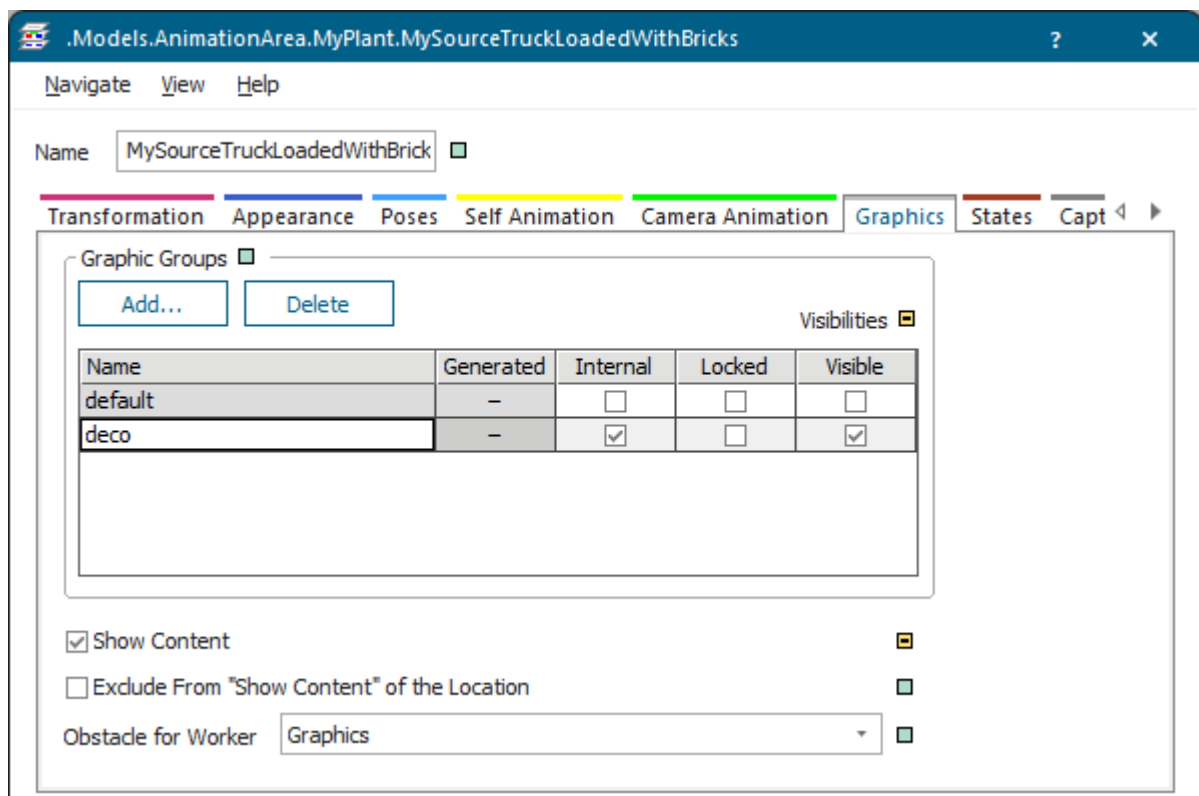
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## Show the Content of the Frame

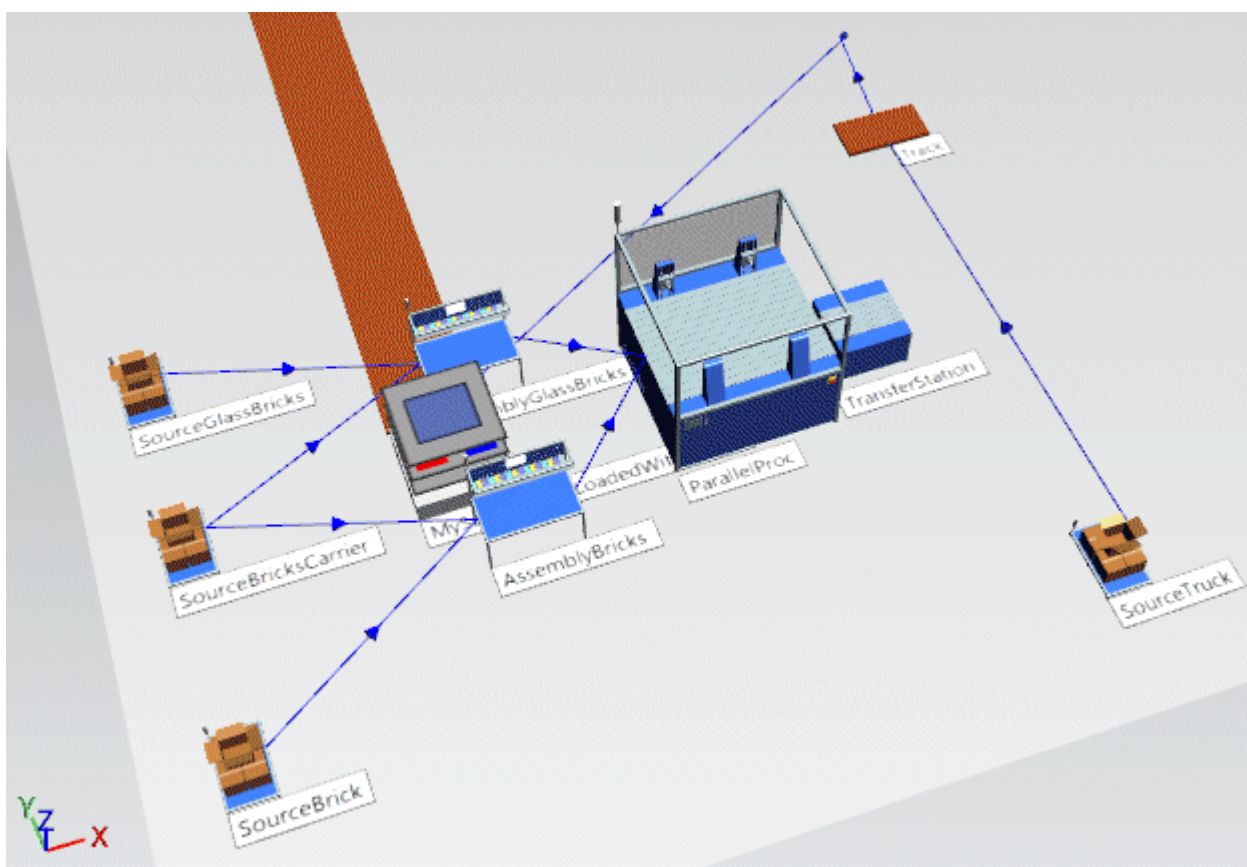
We can also show the content of the *Frame* named *MySourceTruckLoadedWithBricks* in the *Frame* named *MyPlant* instead of the icon of the *Frame*

Proceed as follows:

- Select the *Frame* named *MySourceTruckLoadedWithBricks*. Press the **spacebar**.

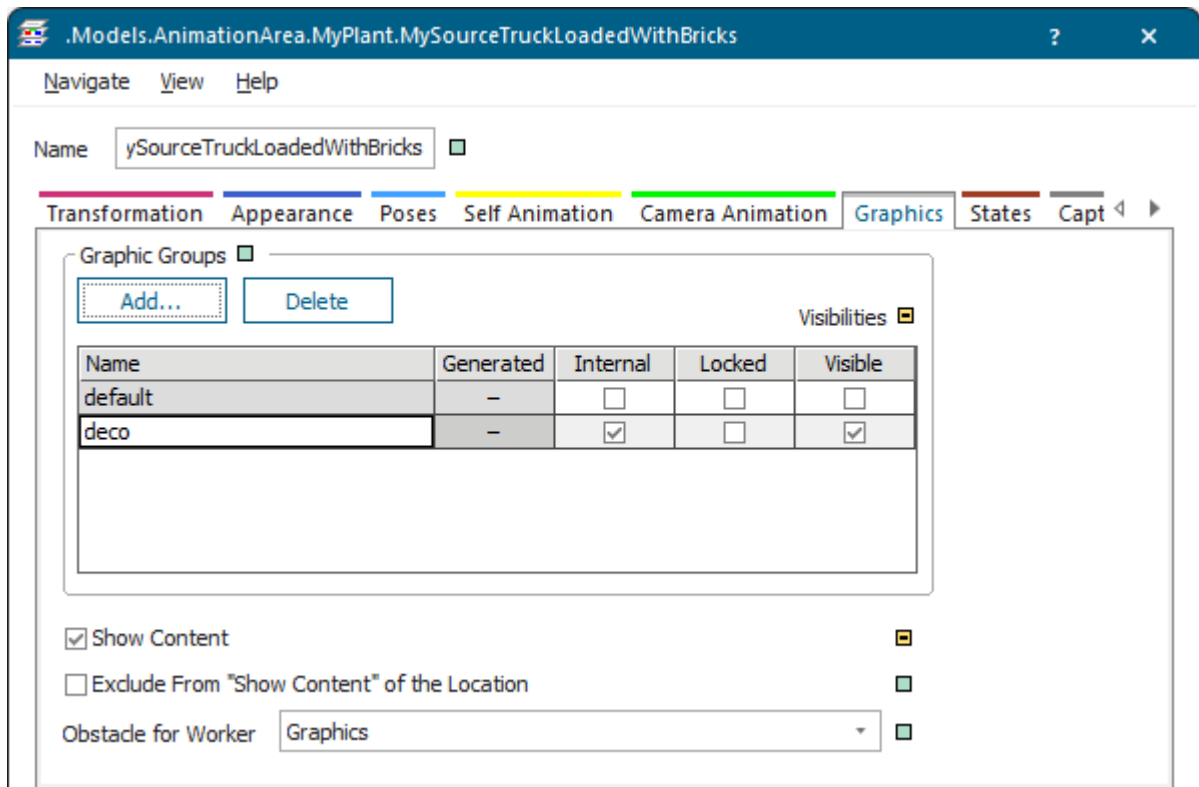


- Select **Show Content**. This then looks like this:



- Adjust the content of the *Frame* named *MySourceTruckLoadedWithBricks*:
  - Clear the check box **Visible** for the graphic group named **default**, so that the default graphic of the *Frame* is not shown over the content of the *Frame*.





- Delete the **Interface** of type **Exit**. We do not need it as our *sub-Frame* only contains a single material flow object that will be connected externally. Connect the *sub-Frame* with the *Track*.
- We finally adjust the positions of the objects and run the simulation: